DIY-QuantiFluorONE-dsDNA-Fluorometer

Workshop Booklet for dsDNA Quantification in eDNA Applications

SHARK-Seq Workshop 2026

S	Stakeholder Engagement
H	Hands-on Exploration of Training Strategies
A	Advancement in Reference Libraries
R	Research Capacity Strengthening
K	Knowledge Sharing
Seq	Sequencing Technology (ONT)

Workshop organizers

Asst. Prof. Dr. Amy Then, Universiti Malaya
Dr. Achim Meyer, LZMT
Dr. Martin Helmkampf, LZMT

Additional Workshop participants

Asst. Prof. Dr. Wanlada Kranunglak, King Mongkut's Institute of Technology Ladkrabang
Dr. Vindi Rilany Manurung, Universitas Sumatera Utara
Dr. Yin Cheong Aden Ip, University of Washington, School of Marine and Environmental Affairs

Booklet author

Dipl.-Ing. Matthias Birkicht; Archived release: https://doi.org/10.5281/zenodo.21909959

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DIY-QuantiFluorONE-dsDNA-Fluorometer Workshop Booklet

Prepared for the SHARK-Seq Project (2026–2027)

eDNA Workshop 2026
with Dr. Achim Meyer

About this booklet

This booklet provides the practical and technical documentation for the DIY-QuantiFluorONE-dsDNA-Fluorometer used in the 2026 eDNA workshop within the SHARK-Seq Project.

The complete project repository, including hardware files, software, calibration tools, validation data, references, and editable documentation, is available on GitHub (Birkicht 2026).

SHARK-Seq contributes to more inclusive ocean observation and demonstrates the potential of on-site sequencing as an introduction to eDNA workflows. Library construction and data processingshare steps with traditional illumina based eDNA workflows while being more robust andbioinformatically lighter, with immediate results. A side aspect is the flowcell reuse strategy whichlikewise benefits biodiversity monitoring in particular in data-poor regions: Short-fragment eDNAamplicons, which rapidly block sequencing pores, are run first; the flowcell is then flushed andreloaded with long-amplicon reference sequences efficiently utilising the remaining ~50-200 pores,substantially reducing costs. The Then lab currently implements traditional eDNA surveys in the same region using MiFishamplicons (Illumina/Sterivex), providing benchmarks for SHARK-Seq. Additionally, Then and Meyersecured ONT funding for whole-genome sequencing of the critically endangered Bornean guitarfish(Rhinobatos borneensis, 20× coverage) on site, creating synergies for consumables and data. SHARK-Seq funds transforms the Then lab into a regional nanopore eDNA hub whilst equipping Thai andIndonesian partners with skills, seeding south–south capacity exchange.This capacity is important as climate-driven range shifts in elasmobranchs necessitate adaptivemonitoring. Rapid, cost-effective eDNA assessment enables responsive governance such as MPA boundary adjustments and fisheries regulations as species distributions change. SHARK-Seq creates a transferable model adaptable to other taxa and geographies, offering a scalable blueprint for inclusive biodiversity monitoring aligned with OBON and Global South priorities.

This booklet covers the theoretical background, hardware and mechanical design, software installation, calibration workflow, measurement protocol, quality-control procedures, analytical validation, troubleshooting, and the information required to reproduce and operate the FluoroQuant instrument (by M. Birkicht) for the dsDNA quantification.

Current content status

This booklet documents the current validated configuration of the DIY-QuantiFluorONE-dsDNA-Fluorometer.

The documented hardware, mechanical components, firmware, calibration workflow, measurement procedure, and analytical validation correspond to the repository version supplied with this booklet.

Any future changes to hardware, firmware, optical components, calibration procedures, or analytical performance should be documented explicitly and evaluated before being treated as equivalent to the configuration described here.

1 Introduction and Learning Objectives

1.1 Purpose

This booklet supports the assembly, software installation, calibration, quality control, and practical use of the DIY-QuantiFluorONE-dsDNA-Fluorometer during the 2026 eDNA workshop within the SHARK-Seq Project (2026–2027).

Project repository: DIY-QuantiFluorONE-dsDNA-Fluorometer

The instrument and its documentation are intended to provide a transparent and reproducible approach to fluorometric dsDNA quantification using the Promega QuantiFluor® ONE dsDNA System. The workshop combines practical instrument construction and operation with calibration, analytical quality control, and interpretation of measurement results.

1.2 Learning objectives

After the workshop, participants should be able to:

- identify the main optical, electronic, and mechanical components of the instrument;
- understand the 90° fluorescence measurement geometry and the roles of the excitation source, optical filters, and TSL2591 detector;
- install and operate the documented PyBadge firmware and navigate the measurement and calibration menus;
- prepare and measure QuantiFluor® ONE dsDNA assays using clean, DNA-free consumables;
- perform reagent-blank and calibration measurements;
- apply two-point or multipoint calibration and recognise the valid working range;
- interpret concentration results together with calibration information, LOD, LOQ, and quality-control criteria;
- document and retain raw measurements, calibration data, and analytical results; and
- understand which hardware, software, optical, or analytical changes require renewed verification or calibration.

1.3 Workshop context

The FluoroQuant-dsDNA-Fluorometer is used as part of the 2026 eDNA workshop with Dr. Achim Meyer within the SHARK-Seq Project.

The practical exercise demonstrates how an openly documented fluorometric instrument can be assembled, calibrated, validated, and used for dsDNA quantification in an eDNA-related laboratory workflow. The emphasis is not only on obtaining a concentration result, but also on understanding the measurement chain and the quality-control information associated with that result.

1.4 Intended use

The FluoroQuant-dsDNA-Fluorometer is a research, teaching, and development instrument intended for laboratory and educational applications.

The documented configuration has been evaluated for the analytical workflow described in this booklet. Changes to the optical configuration, detector settings, firmware, sample geometry, assay procedure, or calibration model may affect analytical performance and should therefore be evaluated before results are considered equivalent to those obtained with the documented configuration.

The instrument is not intended for medical or clinical diagnostic use.

2 Safety and Laboratory Requirements

2.1 General laboratory practice

Use clean, DNA-free consumables and follow the safety documentation supplied with the Promega QuantiFluor® ONE dsDNA System and DNA standards. Wear suitable gloves and laboratory clothing, avoid contamination of open tubes, and dispose of used consumables according to local rules.

2.2 Instrument safety

Keep liquids away from the PyBadge, USB connector, battery, and internal electronics. Switch the instrument off before opening the enclosure or changing hardware. Use only the documented power sources and inspect cables and the LiPo battery before use.

2.3 Workshop responsibility

The local workshop organiser must provide the applicable risk assessment, safety data sheets, emergency information, and waste stream before practical work begins.

3 Theory and References

3.1 Purpose and practical scope

Instrument qualification, assay calibration, method validation, and comparison with established instruments are complementary but distinct elements of analytical quality assurance. Throughout this project, the emphasis is placed on transparent documentation of the complete measurement chain rather than on comparison with a single commercial instrument.

This chapter provides the minimum theory needed to understand, calibrate, and use the DIY-QuantiFluorONE-dsDNA-Fluorometer in a workshop. It is not intended to be a complete textbook on fluorescence spectroscopy or analytical-method validation.

The instrument uses off-the-shelf optical and electronic components, one fluorescence sensor, a fixed 90° optical geometry, and the Promega QuantiFluor® ONE dsDNA assay. Quantitative results are obtained only after blank measurement and calibration. The current repository status and the analytical validation status must always be reported separately.

> **Workshop objective:** Prepare a reagent blank, standards, and unknown samples; measure the blank-corrected fluorescence response; apply a two-point or multipoint calibration; and interpret the result within the calibrated working range.

3.2 Fluorescence in brief

A fluorophore absorbs excitation light and reaches an electronically excited state. Part of the absorbed energy is then emitted as fluorescence at a longer wavelength. The separation between the excitation and emission maxima is the **Stokes shift**.

For the QuantiFluor® ONE dsDNA dye, Promega specifies an excitation maximum of 504 nm and an emission maximum of 531 nm. The nominal Stokes shift is therefore about 27 nm (Promega Corporation 2022).

Within a suitable working range, the fluorescence response increases with the amount of dye-bound dsDNA. The measured signal is also influenced by excitation intensity, filter transmission, sample-vessel geometry, dye binding, sensor settings, temperature, and sample matrix. At high concentrations, detector saturation, inner-filter effects, or dye-response nonlinearity may occur (Lakowicz 2006; Valeur and Berberan-Santos 2012).

Important practical effects are:

- **Background:** signal from the reagent, tube, sensor, stray light, and sample matrix;
- **Quenching:** reduced fluorescence caused by molecular interactions or sample constituents;
- **Photobleaching:** signal loss caused by prolonged illumination;
- **Interference:** fluorescence or absorption from contaminants, extraction reagents, RNA, ssDNA, proteins, salts, or solvents.

These effects are controlled primarily by using a matching reagent blank, identical tubes and geometry, consistent incubation and reading times, and calibration standards prepared in a suitable matrix.

3.3 Optical architecture

The optical path is fixed by the 3D-printed holder:

```
485 nm nominal radial LED
        ↓
Neemoo Ex470BP-40 excitation filter
        ↓
Promega E4941 thin-walled 0.5 mL PCR tube
        ↓
QuantiFluor® ONE dye-dsDNA fluorescence
        ↓  at 90° to the excitation axis
Neemoo Em532BP-40 emission filter
        ↓
one Adafruit TSL2591 fluorescence sensor
```

The TSL2591 sensor axis is positioned **at 90°** to the excitation axis. This geometry reduces direct illumination of the sensor by the excitation beam. The geometry, tube position, filters, LED current, gain, and integration time are part of the calibration and must remain unchanged during a measurement series.

3.3.1 Selection of the Optical Configuration

The optical configuration of the DIY-QuantiFluorONE-dsDNA-Fluorometer was selected to maximize excitation efficiency while minimizing stray excitation light reaching the detector.

The 485 nm excitation LED closely matches the excitation spectrum of the Promega QuantiFluor® ONE dsDNA dye. The Ex470BP-40 excitation filter narrows the LED spectrum, while the Em532BP-40 emission filter efficiently suppresses scattered excitation light and transmits the fluorescence emission around 530 nm.

The spectral relationship between the excitation source, optical filters and fluorescence spectra is illustrated in Figure 3.1.

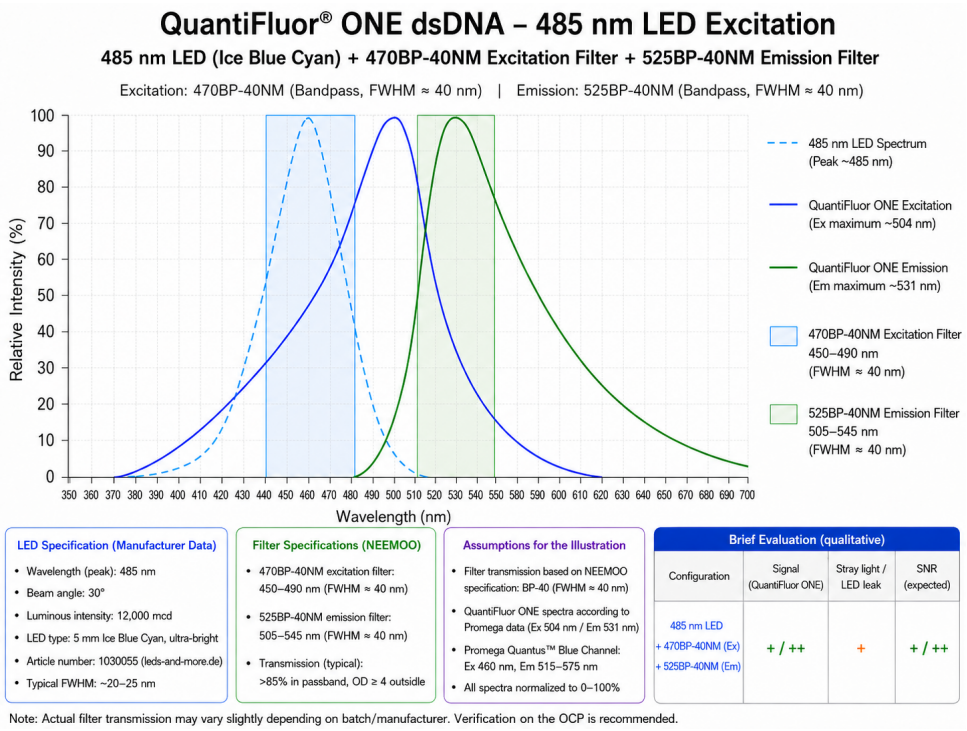

Figure 3.1: Spectral compatibility of the optical configuration

Figure 3.1 Spectral compatibility of the 485 nm excitation LED, Ex470BP-40 excitation filter, Em532BP-40 emission filter and the QuantiFluor® ONE dsDNA fluorescence system. The overlap illustrates the optical basis of the DIY-QuantiFluorONE-dsDNA-Fluorometer.

The same design principle can be applied to other fluorescence assays by selecting appropriate LEDs, optical filters, cuvette holders and calibration models.

3.3.2 Installed optical and electronic components

Component	Installed configuration	Practical role
Excitation LED	5 mm radial 485 nm “Ice Blue/Cyan” LED; 30°; 12,000 mcd; 3.2 V; article 1030055	Excites the dye–dsDNA complex
LED control board	ioRodeo `radial_16mA` fixed-current LED board, `ver_0p1_rev_3`, 16 mA	Provides a defined fixed current for the excitation LED; not an addressable I²C device
Excitation filter	Neemoo Ex470BP-40	Selects the useful blue excitation band and reduces longer-wavelength LED emission
Sample vessel	Promega thin-walled 0.5 mL PCR tube, Cat. No. E4941	Defines the assay volume and optical geometry
Emission filter	Neemoo Em532BP-40	Passes the green fluorescence band and suppresses excitation light
Sensor	Adafruit TSL2591 High Dynamic Range Digital Light Sensor, Product ID 1980, I²C address `0x29`	Measures the filtered fluorescence signal

Component	Installed configuration	Practical role
Multiplexer	Adafruit PCA9546 four-channel STEMMA QT/Qwiic I²C multiplexer, project inventory Product ID 5664, address `0x70`	Selects the connected I²C channels
Controller	Adafruit PyBadge, Product ID 4200	Runs the CircuitPython firmware and displays and stores results
Cabling	STEMMA QT/Qwiic JST SH four-pin cables and SparkFun PRT-15109 Qwiic–Grove adapter	Connects the PyBadge, multiplexer, LED board, and sensor

The PyBadge-based firmware and hardware arrangement build on the open ioRodeo Open Colorimeter Plus ecosystem (ioRodeo 2026b, 2026a). Component identities and connections are documented by the corresponding Adafruit and SparkFun sources (Adafruit Industries n.d.c, n.d.b, n.d.a; SparkFun Electronics n.d.).

3.3.3 Filter verification

The supplier provides the nominal filter designation and percentage-transmittance information (Neemoo n.d.). The actual installed filter specimens were additionally checked with the project's DIY spectrometer based on Leslie Wright's PySpectrometer2 project (Wright 2026).

The project measurements were normalized to the maximum of each curve and were used to verify the passband position and shape:

- Ex470BP-40: approximate half-maximum interval 450–490 nm;
- Em532BP-40: approximate half-maximum interval 518–555 nm.

The measurements confirm the intended spectral separation but are not used as an absolute calibration of filter transmittance.

3.4 Signal formation and blank correction

The TSL2591 provides a full-spectrum channel and an infrared channel. The firmware calculates a project-defined visible signal for each reading:

$$VIS_i = \max(FULL_i - IR_i, 0)$$

One measurement cycle consists of three technical sensor readings. Their mean is used as the cycle result:

$$\overline{VIS} = \frac{1}{3}\sum_{i=1}^{3} VIS_i$$

The reagent blank contains the QuantiFluor® ONE reagent but no added dsDNA. The blank-corrected response is:

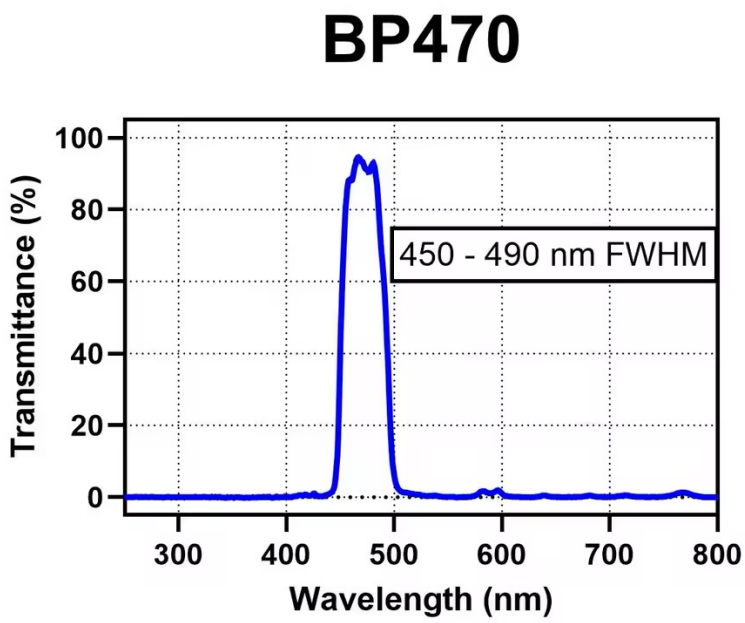

Figure 3.2: Spectral verification of the installed Neemoo Ex470BP-40 excitation filter. The project measurement was normalized to its maximum; the approximate half-maximum interval was 450–490 nm.

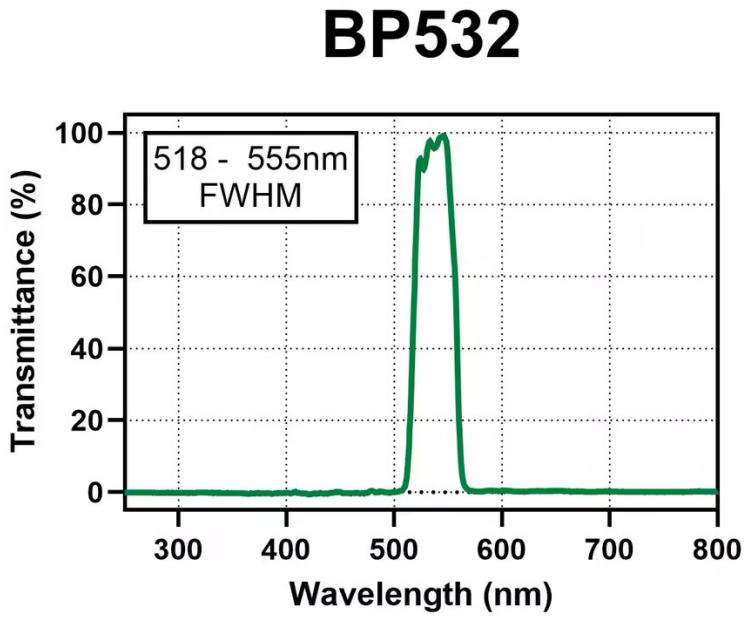

Figure 3.3: Spectral verification of the installed Neemoo Em532BP-40 emission filter. The project measurement was normalized to its maximum; the approximate half-maximum interval was 518–555 nm.

$$RFU = \overline{VIS}_{sample} - \overline{VIS}_{blank}$$

RFU means **relative fluorescence unit**. It is an instrument-dependent response, not an SI unit. Signed RFU values are retained in the raw data. A negative value indicates that the measured sample response was below the stored blank response; it must not be interpreted as a negative physical DNA concentration.

A new blank should be measured when the reagent lot, tube type, optical arrangement, sensor settings, temperature conditions, or measurement session changes.

3.5 dsDNA quantification with QuantiFluor® ONE

The QuantiFluor® ONE dsDNA System is an add-and-read fluorescent assay for purified dsDNA. Promega specifies 504 nm excitation, 531 nm emission, and a nominal range of 0.2–400 ng dsDNA input. The kit includes QuantiFluor® ONE Lambda DNA at 400 µg/mL as a calibration material (Promega Corporation 2022).

The project uses:

```
1 µL sample or standard + 200 µL QuantiFluor® ONE reagent
= 201 µL physical total volume
```

Promega recommends thin-walled 0.5 mL PCR tubes, thorough mixing without bubbles, five minutes of incubation at room temperature protected from light, and careful pipetting of the small sample volume (Promega Corporation 2022).

3.5.1 λ-DNA standards

Promega supplies λ-DNA as the dsDNA calibration material for the QuantiFluor® ONE assay. The stock concentration specified by the manufacturer is 400 µg/mL, which is exactly equivalent to 400 ng/µL. Because the fluorometer reports DNA concentrations in ng/µL and the assay range is expressed in the same unit, 400 ng/µL is used throughout this booklet for consistency (Promega Corporation 2022).

Record the supplier, product, lot, stock concentration, dilution medium, preparation date, and storage conditions of all calibration standards. Use fresh low-concentration dilutions where practical. Promega recommends thorough mixing of the standard and assay reagent; vortex well while avoiding persistent air bubbles.

Promega also notes that a dsDNA standard with a molecular size similar to that of the unknown DNA can improve comparability. Therefore, λ-DNA is a practical and well-defined workshop calibration standard, but differences in DNA fragment size and sample matrix may remain sources of bias (Promega Corporation 2022).

Lambda DNA is a linear dsDNA molecule containing 48,502 bp. Using an approximate average molecular mass of 650 Da per base pair, intact λ-DNA has a molecular mass of approximately 31.5 MDa. Thus, 1 ng of intact λ-DNA corresponds to approximately 0.032 fmol, or about 32 fmol per µg.

The DNA mass contained in a selected volume is calculated from its mass concentration:

$$m\ (\text{ng}) = c\ (\text{ng}/\mu\text{L}) \times V\,(\mu\text{L})$$

The corresponding molar amount can then be estimated from the DNA mass and the representative fragment length:

$$n_{\mathrm{DNA}}\ (\mathrm{fmol}) \approx \frac{m\ (\mathrm{ng}) \times 10^6}{650 \times L\ (\mathrm{bp})}$$

Combining both steps gives the direct conversion from a measured DNA concentration:

$$n_{\mathrm{DNA}}\ (\mathrm{fmol}) \approx \frac{c\ (\mathrm{ng}/\mu\mathrm{L}) \times V\,(\mu\mathrm{L}) \times 10^6}{650 \times L\ (\mathrm{bp})}$$

where c is the DNA concentration in ng/µL, V is the selected DNA volume in µL, and L is the representative DNA fragment length in base pairs.

For the supplied Promega λ-DNA standard, 1 µL of the 400 ng/µL stock contains:

$$m = 400\ \mathrm{ng}/\mu\mathrm{L} \times 1\ \mu\mathrm{L} = 400\ \mathrm{ng}$$

For intact 48,502-bp λ-DNA:

$$n_{\mathrm{DNA}} \approx \frac{400 \times 10^6}{650 \times 48{,}502} \approx 12.7\ \mathrm{fmol}$$

Thus, 1 µL of the 400 ng/µL λ-DNA stock contains 400 ng DNA but only approximately 12.7 fmol of intact λ-DNA molecules. The same DNA mass represents a much larger molar amount when the DNA is present as shorter fragments.

PCR context: In the downstream PCR workflow used for this workshop, approximately 100 fmol DNA template is targeted, with 80 fmol regarded as the minimum acceptable input. This requirement cannot be inferred directly from the fluorometrically measured concentration in ng/µL: conversion to fmol additionally requires the volume of DNA used and an appropriate representative fragment length.

For example, 100 fmol corresponds to approximately 9.8 ng at 150 bp, 32.5 ng at 500 bp, and 65 ng at 1000 bp.

Getting started with instrument operation, use of the Calibration Suite, sample-list editing, and the dsDNA measurement protocol is covered in Chapters 6–8.

3.6 Analytical validity, reference methods, and open measurement chains

Instrument qualification, assay calibration, and method validation are related but distinct tasks. A fluorometer can be checked for properties such as linear response, wavelength suitability, detector response, sensitivity, and day-to-day stability without treating another fluorometer as an absolute reference. General fluorescence-quality guidance likewise distinguishes instrument characterization and performance verification from the validation of the analytical method itself.[1]

[1] See Resch-Genger U, Hoffmann K, Nietfeld W, et al. *How to improve quality assurance in fluorometry: fluorescence-inherent sources of error and suited fluorescence standards.* Journal of Fluorescence. 2005;15:337–362. doi:10.1007/s10895-005-2630-3; and DeRose PC, Resch-Genger U. *Recommendations for Fluorescence Instrument Qualification: The New ASTM Standard Guide.* Analytical Chemistry. 2010;82(5). These works discuss instrument qualification, linearity, spectral response, wavelength accuracy, sensitivity, and performance verification as elements of quantitative fluorescence quality assurance.

For dsDNA quantification, a commercial fluorometer is therefore useful as a **comparison instrument**, but agreement with one commercial instrument is not the only possible evidence of analytical validity. A stronger validation strategy combines several independent lines of evidence: assigned or independently characterized DNA standards, repeatability and linearity studies, blank behaviour, recovery, LOD/LOQ, and comparison with an analytically independent method.

For sufficiently pure dsDNA at concentrations suitable for absorbance measurement, UV photometry at 260 nm provides such an independent comparison principle through the Beer–Lambert relationship and an accepted DNA-specific mass-extinction convention. Fluorometric and photometric DNA determinations are not chemically identical measurements: absorbance at 260 nm responds to total UV-absorbing nucleic-acid material and other absorbing contaminants, whereas the QuantiFluor® ONE assay is designed to respond preferentially to dsDNA. Promega therefore notes that concentrations obtained by fluorescence and by optical-density measurements may differ because the two methods measure different properties of the sample (Promega Corporation 2022). Agreement should consequently be evaluated on clean, well-characterized dsDNA standards rather than assumed for arbitrary biological extracts.

This distinction is particularly useful for open-science instrumentation. The analytical evidence need not form a hierarchy in which an open instrument is considered valid only because it reproduces the result of a proprietary instrument. Instead, the DIY-QuantiFluorONE can be evaluated within an **open measurement chain**: reference or assigned DNA materials define the sample, photometric determination provides an independent physical-chemical comparison where applicable, open fluorometers such as DIYNAFLUOR provide reproducible peer systems, and established commercial instruments provide continuity with routine laboratory practice.

The objective is therefore not to designate one instrument as the universal reference, but to demonstrate that the complete measurement procedure is transparent, reproducible, and consistent with independent analytical evidence within stated uncertainty and applicability limits. This is also why the project retains raw signals, calibration metadata, blank statistics, model diagnostics, and validation data rather than reporting only a final concentration.

3.7 Calibration

Calibration links the blank-corrected fluorescence response to an assigned dsDNA concentration or DNA mass. The linear model used by the project is:

$$y = a + bx$$

where y is RFU, x is the selected concentration or mass basis, a is the intercept, and b is the slope. For an unknown sample:

$$\hat{x} = \frac{y - a}{b}$$

The calibration basis and unit must be stored with the calibration. The firmware normally reports the original-sample concentration in ng/µL and also records the corresponding in-assay concentration.

3.7.1 Two-point calibration

The two-point workflow uses a reagent blank and one known standard. After blank correction, the working intercept is zero:

$$b = \frac{RFU_{standard}}{x_{standard}}$$

$$x_{sample} = \frac{RFU_{sample}}{b}$$

Two-point calibration is fast and suitable for a workshop routine when the assay response is known to be linear in the selected range. It does not by itself test linearity or provide a residual-based assessment across the range.

3.7.2 Multipoint calibration

Multipoint calibration uses several concentration levels and replicate measurements. Calibration Suite 7.3 can calculate ordinary least squares (OLS) and compare it with weighted models (`1/c`, `1/c²`, and `1/s²`). It also displays residuals, confidence bands, prediction bands, recovery, and model-comparison statistics.

The current PyBadge firmware imports accepted OLS calibration data. Model selection and diagnostics are performed in Calibration Suite 7.3 before the calibration JSON is transferred to the instrument.

DIN 38402-51:2017-05 is used as a reference for linear calibration concepts. The repository cites the standard but does not reproduce protected standard text (*DIN 38402-51:2017-05: German Standard Methods for the Examination of Water, Waste Water and Sludge–General Information–Part 51: Calibration of Analytical Methods–Linear Calibration (a 51)* 2017).

> **Practical rule:** Do not report a sample quantitatively outside the calibrated range. Dilute or re-prepare the sample and measure it again.

3.8 LOD and LOQ

The limit of detection (LOD) and limit of quantification (LOQ) depend on the selected procedure, calibration, blank variability, and dataset. They are not fixed properties of the dye or sensor alone.

3.8.1 Two-point firmware estimate

For an active two-point calibration, the firmware can measure ten independently prepared reagent blanks. Each blank is measured as one complete three-reading cycle. The sample standard deviation of the ten blank-cycle means is used:

$$LOD_{estimate} = \frac{3s_{blank}}{b}$$

$$LOQ_{estimate} = \frac{10s_{blank}}{b}$$

These are transparent blank-based estimates for workshop and instrument evaluation. They are not presented as a complete implementation of every procedure in DIN 32645.

3.8.2 Multipoint calculation in Calibration Suite 7.3

For multipoint calibration, Calibration Suite 7.3 implements the indirect calibration-curve calculation documented in the software as the DIN 32645 method summarized in CLB1. The calculation uses the residual standard deviation, slope, Student-t factors, calibration design, significance level, the selected quantification factor k, and the number of future determinations. The LOQ is solved iteratively.

DIN 32645:2008-11 is cited as the relevant standard for decision, detection, and quantification limits under repeatability conditions (*DIN 32645:2008-11: Chemical Analysis–Decision Limit, Detection Limit and Determination Limit Under Repeatability Conditions–Terms, Methods, Evaluation* 2008). A formal claim of conformity requires a documented comparison of the software implementation with the licensed standard and the defined validation protocol.

Always report LOD and LOQ together with the calibration ID, method, unit, significance level, k factor, number of calibration levels and replicates, and software version.

3.9 Reference projects and commercial comparison

System	Relevant role in this project	Main distinction
ioRodeo Open Colorimeter Plus	Open hardware and CircuitPython reference platform	General open optical-instrument platform used as a technical base
DIYNAFLUOR	Open-source reference for an inexpensive 90° TSL2591 nucleic-acid fluorometer	Developed primarily for Qubit assays; uses two-point calibration and off-the-shelf components
DIY-QuantiFluorONE	Workshop instrument described here	QuantiFluor® ONE assay, Ex470BP-40, Em532BP-40, one TSL2591 at 90°, two-point and imported multipoint calibration
Promega Quantus™	Commercial reference instrument	Blue excitation peak 470 nm, blue emission band 510–580 nm, solid-state detector, and manufacturer-defined calibration workflow

DIYNAFLUOR demonstrates that a practical nucleic-acid fluorometer can be constructed from readily available parts, a 3D-printed optical head, a 90° optical path, and a TSL2591 sensor (Anderson et al. 2024; Trau Lab 2026). The DIY-QuantiFluorONE design follows the same practical open-hardware philosophy but uses a different controller, LED board, filter set, assay, and calibration workflow.

Promega specifies the Quantus™ blue channel with a 470 nm excitation peak, excitation wavelengths up to 495 nm, and an emission range of 510–580 nm (Promega Corporation 2024). These ranges are compatible with the QuantiFluor® ONE assay and provide a useful commercial reference. Direct statements about bias, precision, or agreement require measurements of the same standards on both instruments.

3.10 Practical validity conditions and limitations

A calibration remains valid only while the measurement configuration remains unchanged. Recalibrate or verify the calibration after changing any of the following:

- LED, LED current, excitation filter, or emission filter;
- tube type, holder, insertion depth, or optical alignment;
- TSL2591 gain, integration time, sensor board, or multiplexer channel;
- reagent formulation or lot;
- sample and reagent volumes;
- concentration basis or blank preparation.

During workshop measurements:

1. Use the same tube type for blank, standards, and unknowns.
2. Keep tube orientation and insertion depth consistent.
3. Avoid bubbles, droplets above the liquid level, fingerprints, and scratches.
4. Mix and incubate all tubes consistently and protect them from light.
5. Measure a reagent blank before samples.
6. Use standards that bracket the expected sample concentration.
7. Repeat or dilute samples outside the calibrated range.
8. Retain raw FULL, IR, VIS, blank, RFU, calibration ID, and result data.

The method is intended for research, teaching, and workshop use. It is not a clinical diagnostic method. The supplier's nominal assay range is not automatically the validated working range of the assembled DIY instrument.

3.11 Workshop take-away

The practical measurement chain is:

```
Prepare blank and standards
→ incubate consistently
→ measure one TSL2591 at 90°
→ calculate VIS and blank-corrected RFU
→ apply the selected calibration
→ check range, LOD/LOQ, and validity conditions
→ save the raw and calculated data
```

The practical implementation of these analytical concepts, including calibration acceptance, quality control, uncertainty evaluation, and routine decision criteria, is described in Chapter 9.

4 Hardware and Assembly

4.1 Purpose and workshop outcome

This chapter guides participants through the identification, assembly, inspection, and first hardware check of the DIY-QuantiFluorONE-dsDNA-Fluorometer. It is intended as a practical build guide rather than a comprehensive electronics or fluorescence textbook.

At the end of the workshop, the participant should be able to:

- identify every electronic, optical, and mechanical component;
- install the 485 nm excitation source and both optical filters;
- install the Promega E4941 tube holder and the single TSL2591 sensor;
- confirm that the detection axis is fixed at 90° to the excitation axis;
- connect the PyBadge, PCA9546 multiplexer, TSL2591, and fixed-current LED board;
- perform a pre-power inspection and a first hardware test;
- locate the source, CAD, PCB, Gerber, and assembly files needed to reproduce the instrument.

The instrument uses **one TSL2591 sensor only**. It does not use a second reference or transmission sensor.

Figure 4.1: Assembled DIY-QuantiFluorONE prototype during a firmware and display check. The displayed measurement is not presented as analytical validation data.

4.2 Complete component overview and bill of materials

The machine-readable bill of materials is stored in `bom/master_bom.csv`. The core configuration is summarized below.

Group	Component	Confirmed configuration	Quantity
Controller	Adafruit PyBadge	Product ID 4200	1
I²C distribution	Adafruit PCA9546 multiplexer	Product ID 5664; address `0x70`	1
Detector	Adafruit TSL2591 STEMMA QT	Product ID 1980; address `0x29`	1
LED driver	ioRodeo fixed-current radial LED board	`radial_16mA`, `ver_0p1_rev_3`	1
Excitation source	5 mm Ice Blue/Cyan radial LED	485 nm; 30°; 12,000 mcd; 3.2 V; article 1030055	1
Excitation filter	Neemoo Ex470BP-40	**8 x 8 x 1 mm**, installed	1
Emission filter	Neemoo Em532BP-40	**8 x 8 x 1 mm**, installed	1
Sample vessel	Promega E4941 thin-walled 0.5 mL PCR tube	workshop sample tube	as required
Cabling	STEMMA QT / Qwiic JST SH four-pin cables	10 cm	1
Cabling	STEMMA QT / Qwiic JST SH four-pin cables	5 cm	1
Adapter	SparkFun Qwiic-to-Grove adapter cable	PRT-15109	1
Mechanical assembly	3D-printed PyBadge enclosure	STEP files included	1 set
Optical assembly	3D-printed tube and optics holder	step file	1
Light control	3D-printed light shield	step file	1
Programming and power	USB Micro cable	programming, power, and charging	1
Portable power	Powerbank	10000 mAh	1
Programming	Laptop	with Thonny + USB-Port	1
Portable power	3.7 V LiPo battery	1200 mAh; status optional	0 or 1

The confirmed mounting fasteners are:

- four M2.5 x 20 screws with matching M2.5 nuts for fastening the PyBadge cover to the enclosure;
- four M3 x 20 screws with matching M3 nuts for fastening the PCR-tube holder to the enclosure.
- two M2 x 6 screws with matching nuts for emission filter holder C

- two M3 x 40 screws with matching M3 nuts and washers for fastening the cuvette holder to the led-cover
- two M3 x 35 screws with matching M3 nuts for fastening the cuvette holder

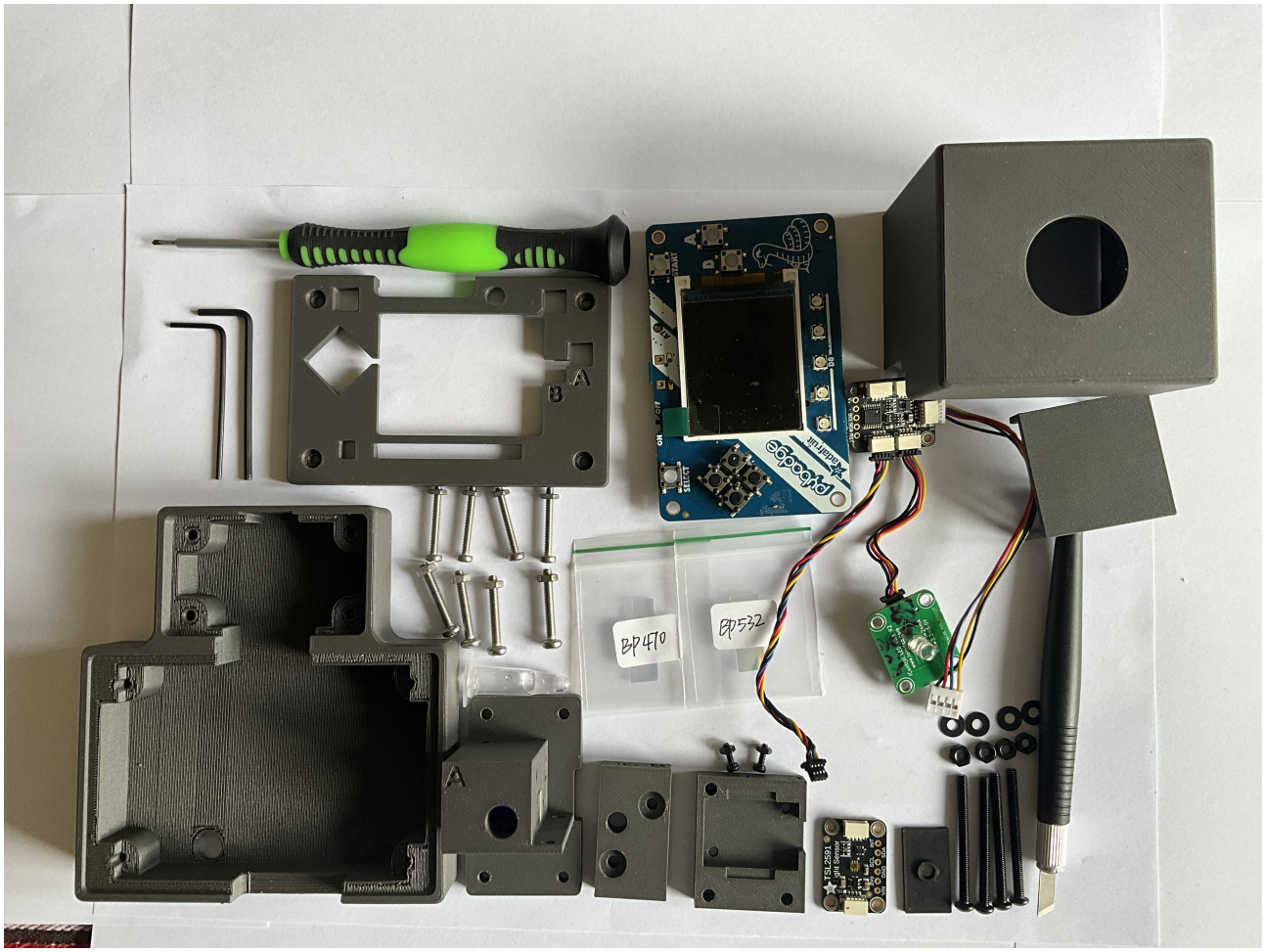

Figure 4.2: Main electronic, optical, mechanical, and laboratory components used to assemble and operate the DIY-QuantiFluorONE-dsDNA-Fluorometer.

4.3 Function of the components

4.3.1 Electronic components

Adafruit PyBadge. The PyBadge is the controller and user interface. It provides the display, buttons, reset switch, on/off switch, USB Micro connection, LiPo connector, battery charging circuit, and the I²C interface used by the external modules. The documented software baseline for this project is CircuitPython 9.1.1.

PCA9546 multiplexer. The multiplexer divides the upstream I²C connection into four selectable downstream branches. Its address is `0x70`. The firmware scans the available multiplexer channels for the TSL2591 sensor, so a fixed downstream channel assignment is not required.

TSL2591 sensor. The TSL2591 detects light passing through the emission filter. Its address is `0x29`. Only one TSL2591 is installed, and its detection axis is mechanically fixed at 90° to the excitation axis.

ioRodeo fixed-current LED board. Revision `ver_0p1_rev_3` supplies a fixed nominal LED current of 16 mA. It is not an addressable I²C device. Its SDA and SCL conductors are pass-through connections, so the board must not be expected to appear in an I²C scan.

4.3.2 Optical components

485 nm LED. The LED provides excitation light. Its radial package must be installed with the correct polarity and must be seated without bending the leads against the printed holder.

Ex470BP-40 excitation filter. This **8 x 8 x 1 mm** filter is positioned between the LED and sample tube. It limits the excitation spectrum before the light reaches the sample.

Em532BP-40 emission filter. This **8 x 8 x 1 mm** filter is positioned between the sample tube and TSL2591. It reduces direct excitation light and passes the fluorescence region used by the detector.

Promega E4941 tube. The thin-walled 0.5 mL PCR tube defines the sample container used by the mechanical holder. The tube must reach the mechanical stop reproducibly and must not be forced into the holder.

4.3.3 Mechanical components

The printed parts perform four critical tasks:

1. locate the sample tube at a reproducible depth;
2. align the LED and excitation filter with the excitation axis;
3. align the emission filter and active TSL2591 area with the detection axis;
4. shield the optical path from room light and display light.

The holder establishes the 90° geometry mechanically. The mechanical development of the optical sample holder also draws on the open ioRodeo fluorometer tube-holder project. The original repository provides FreeCAD design files for the fluorometer tube holder and is distributed under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence.

The DIY-QuantiFluorONE holder combines and adapts elements of these open designs for the Promega E4941 thin-walled 0.5 mL PCR tube, the installed $8 \times 8 \times 1$ mm optical filters, one TSL2591 sensor, and the project-specific 90° optical geometry (Anderson et al. 2024; Trau Lab 2026) (IO Rodeo n.d.)

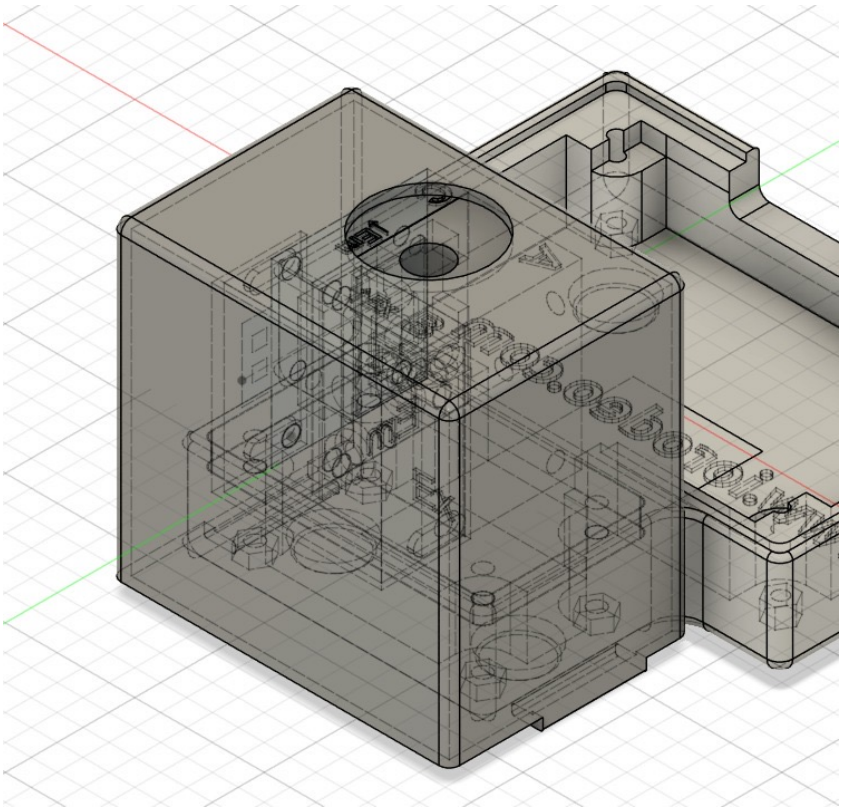

Figure 4.3: CAD overview of the PyBadge enclosure and instrument base.

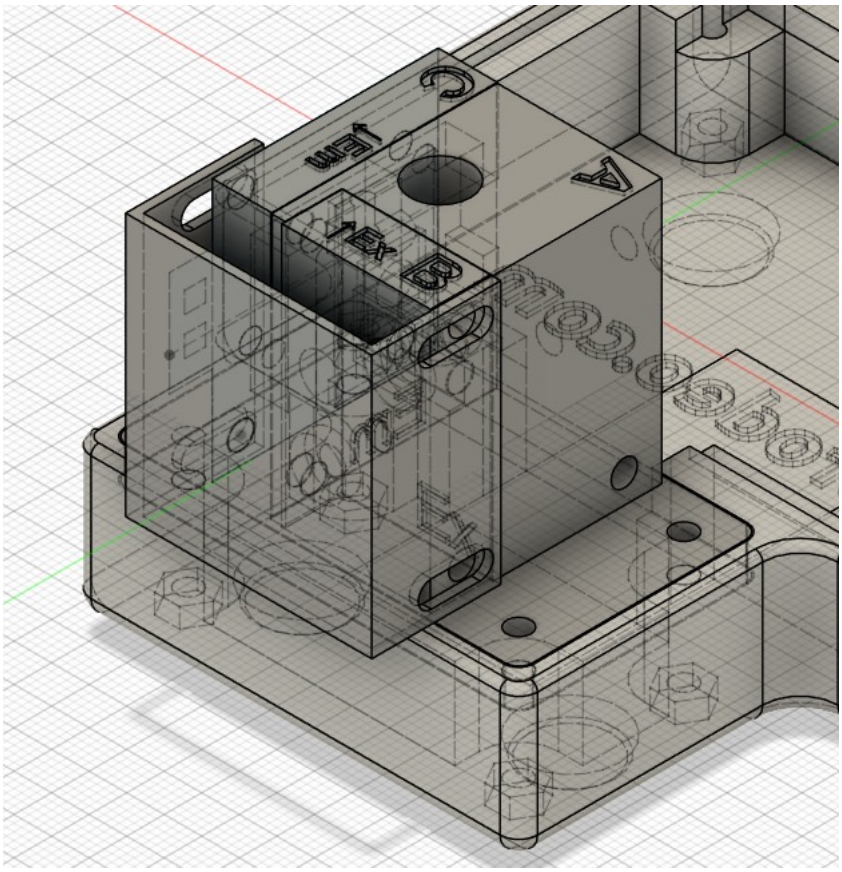

Figure 4.4: CAD overview of the optical module mounted on the instrument base.

4.4 Preparation and inspection of the 3D-printed parts

Before installing electronics or filters:

1. Compare every printed part with the CAD screenshots and file names.
2. Remove support material completely from tube, LED, filter, sensor, screw, and cable openings.
3. Check that no loose strands, curled edges, or partially detached layers can enter the optical path.
4. Inspect the tube bore for ridges that could scratch or jam an E4941 tube.
5. Insert an empty tube gently and confirm that it reaches the stop without excessive force.
6. Check the filter pockets with a clean **8 x 8 x 1 mm** test piece where possible.
7. Check that the LED board and TSL2591 can be inserted without bending the PCBs.
8. Hold the empty printed assembly against a bright lamp and mark any visible light leaks.
9. Clean all printed debris before bringing the optical filters near the holder.

Do not enlarge filter pockets with a power tool while the filters or electronics are installed. Fine manual correction is safer and easier to control.

4.5 Installation of the LED and excitation filter

1. Disconnect USB and remove the LiPo battery if fitted.
2. Identify the LED anode and cathode from the component data and PCB markings.
3. Insert the 5 mm 485 nm LED into the ioRodeo radial LED board in the documented polarity.
4. Confirm that the LED body is straight and that its optical axis points through the excitation opening.
5. Solder only after checking orientation. Avoid prolonged heating of the LED leads.
6. Trim leads so that they cannot touch the enclosure or another conductor.
7. Handle the **8 x 8 x 1 mm** Ex470BP-40 by its edges. Do not touch the clear aperture.
8. Insert the filter into the excitation-filter pocket without twisting or forcing it.
9. Confirm that the filter fully covers the optical opening and cannot rattle into the tube chamber.
10. Install the Part B with two M3 x 40 screws and nuts
11. Install the 485 nm LED-board and the holder. Fix ist with 3M x 40 screws and nuts.

    Check: Looking from the LED side, the sequence must be LED → Ex470BP-40 → sample tube.

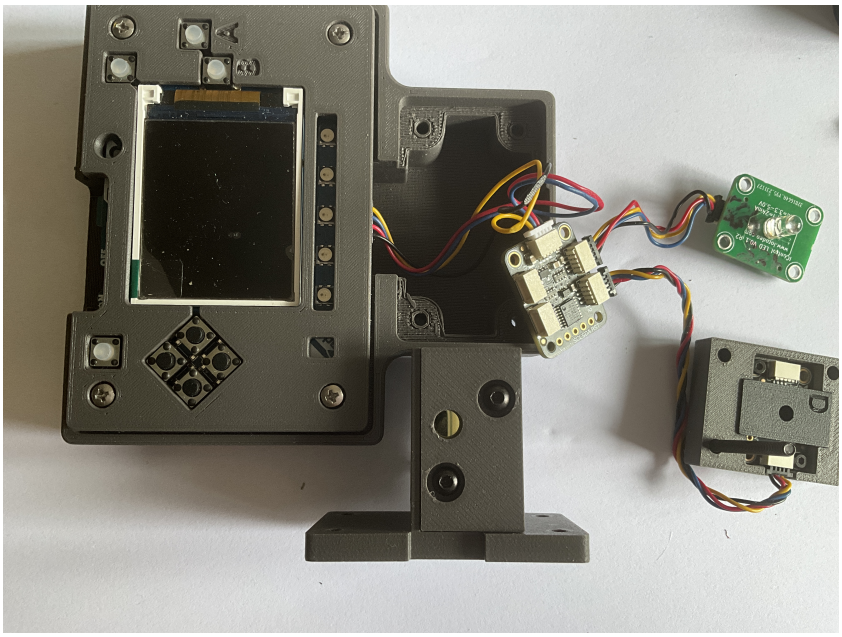

Figure 4.5: Excitation side of the optical module showing PartC with the **installed** 485 nm LED and Ex470BP-40 excitation filter.

4.6 Installation of the Promega E4941 sample-tube holder

1. The E4941 tube must enter vertically and reach a defined mechanical stop.
2. The tube must not tilt toward either optical opening.
3. The tube wall must not touch the LED or sensor board.
4. The optical interrogation region must remain at the same height for every tube.
5. The tube must be removable without pulling the optical module or cables.
6. The holder must block stray light around the upper tube opening when the light shield is installed.

4.7 Installation of the emission filter and single TSL2591 sensor

1. Handle the **8 x 8 x 1 mm** Em532BP-40 only by its edges.
2. Insert it into the detection-side filter pocket.
3. Confirm that it fully covers the detection opening.
4. Identify the active optical area of the TSL2591 board.
5. Orient the board so that the active area faces the emission filter directly.
6. Seat the board flat against its locating surfaces.
7. Fasten it without bowing the PCB.
8. Route the Qwiic cable away from the sample tube and optical openings.
9. Confirm once more that only one TSL2591 is present.

Check: Looking from the detector side, the sequence must be sample tube → Em532BP-40 → TSL2591.

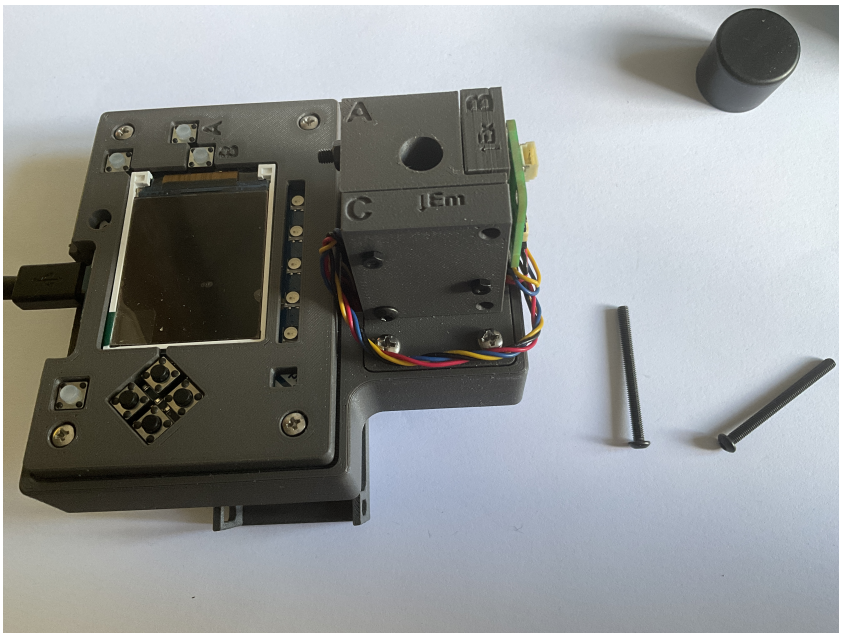

Figure 4.6: Detection side of the optical module showing the Em532BP-40 emission filter and TSL2591 sensor.

4.8 Correct 90° optical geometry

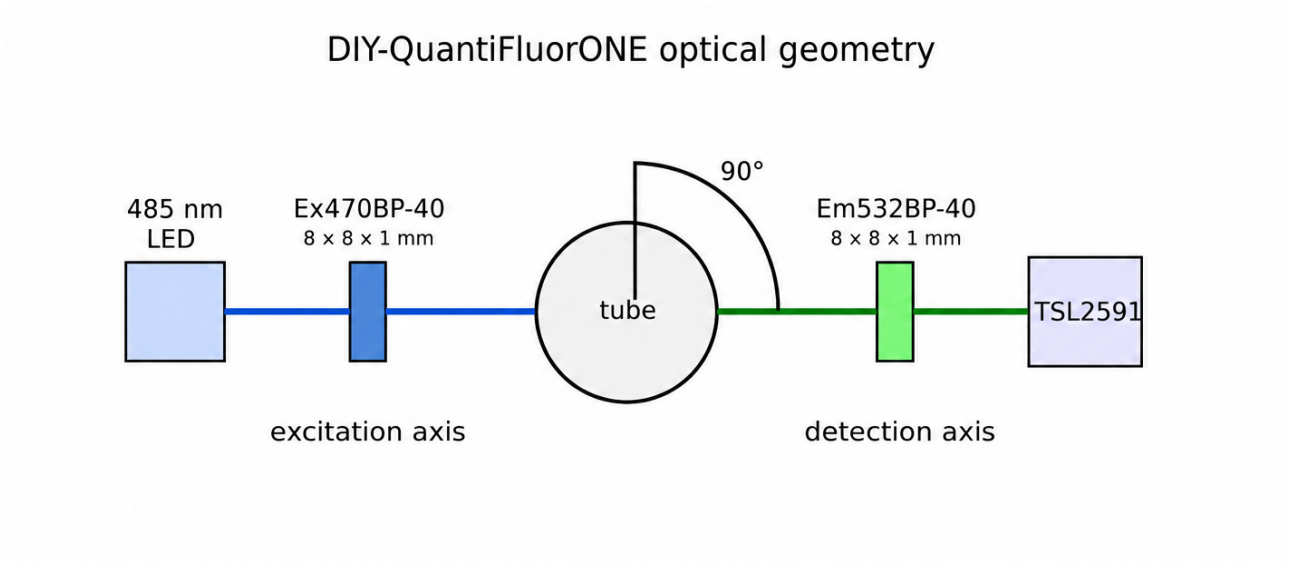

Figure 4.7: Optical geometry of the instrument.

The excitation axis passes from the LED through the Ex470BP-40 to the sample. The detection axis passes from the sample through the Em532BP-40 to the single TSL2591. The printed holder fixes these axes at 90°.

The workshop inspection is mechanical and visual:

- the LED aperture must point to the tube centerline;
- the detector aperture must point to the same tube region;
- the two axes must be orthogonal in the CAD and printed locating features;

- the tube must reach the designed stop;
- neither filter may be tilted, displaced, or partly outside its aperture;
- the TSL2591 active area must face the emission-filter opening.

Do not compensate for poor printed fit by angling the sensor board or LED. Correct the printed part or replace it.

4.9 Installation of the PyBadge, multiplexer, and LED board

The excitation LED is driven by the ioRodeo `radial_16mA` fixed-current LED board, revision `ver_0p1_rev_3`. The component-placement drawing below identifies the principal components and provides a visual reference for inspection of the assembled board.

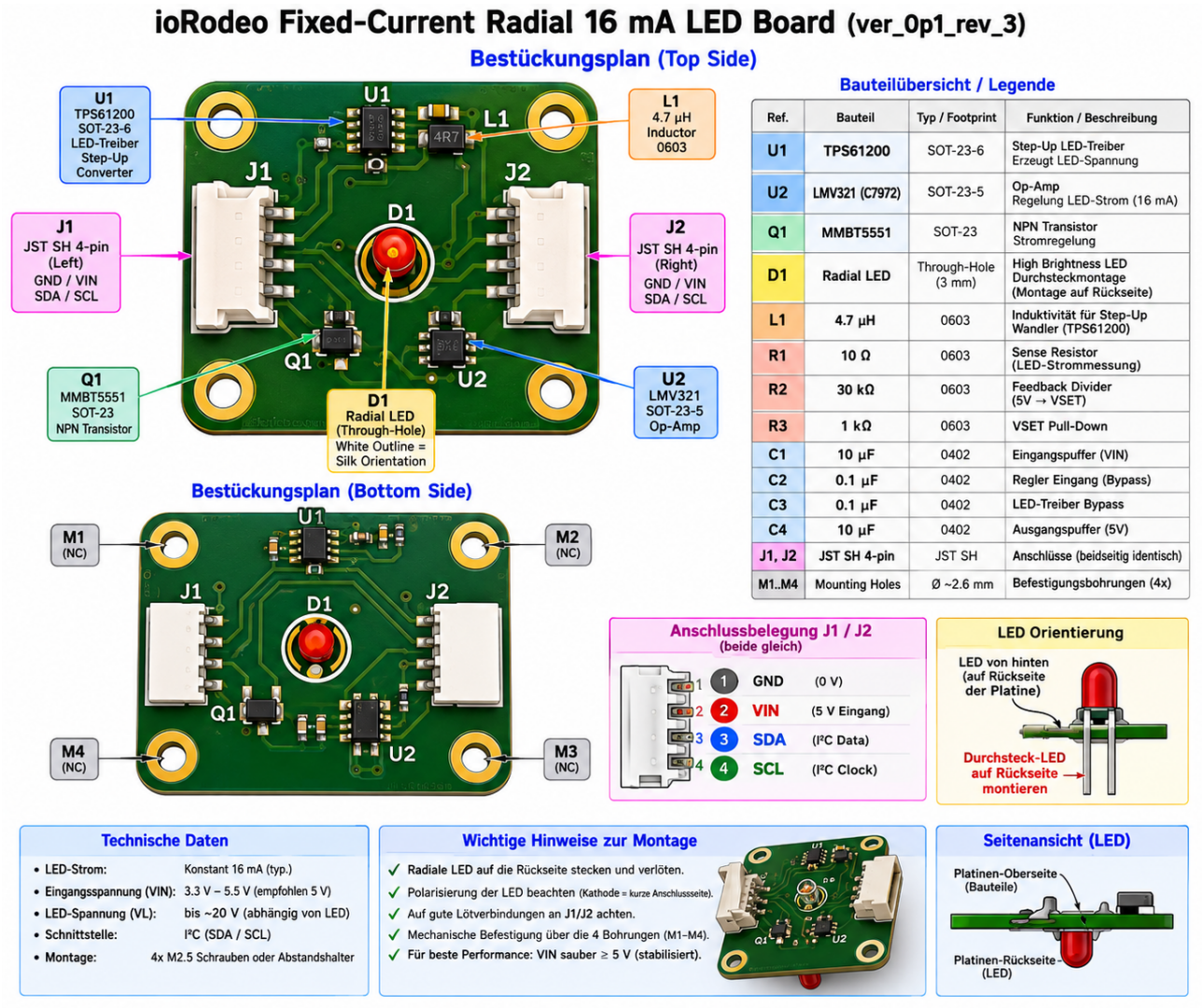

Figure 4.8: Component placement of the ioRodeo `radial_16mA` fixed-current LED board, revision `ver_0p1_rev_3`. The drawing identifies the principal components used for visual inspection and assembly verification.

The principal components are:

- **U1** – TPS61222DCKR boost converter, fixed 5 V;
- **U2** – LMV321 operational amplifier, SOT-23-5;

- **Q1** – MMBT5551 NPN transistor;
- **L1** – 4.7 µH inductor;
- **C1, C2, C4** – 10 µF ceramic capacitors;
- **C3** – 100 nF ceramic capacitor;
- **R1** – 10 Ω resistor;
- **R2** – 30 kΩ resistor;
- **R3** – 1 kΩ resistor;
- **J1, J2** – four-pin JST-SH connectors; and
- **D1** – radial excitation LED position.

The drawing is intended as an assembly and visual-inspection aid.

1. Place the PyBadge in the enclosure without connecting a battery.
2. Confirm access to the reset button, on/off switch, USB Micro port, and LiPo connector.
3. Fit the cover and check that no button is held down by the printed part.
4. Install the PCA9546 in its intended mounting position.
5. Install the ioRodeo `radial_16mA` board near the optical module without allowing its solder joints to touch the enclosure hardware.
6. Ensure that the LED board revision is `ver_0p1_rev_3`.
7. Keep all boards mechanically supported. Qwiic cables must not be used as structural restraints.
8. Leave enough service loop to disconnect a board without pulling directly on the wire bundle.

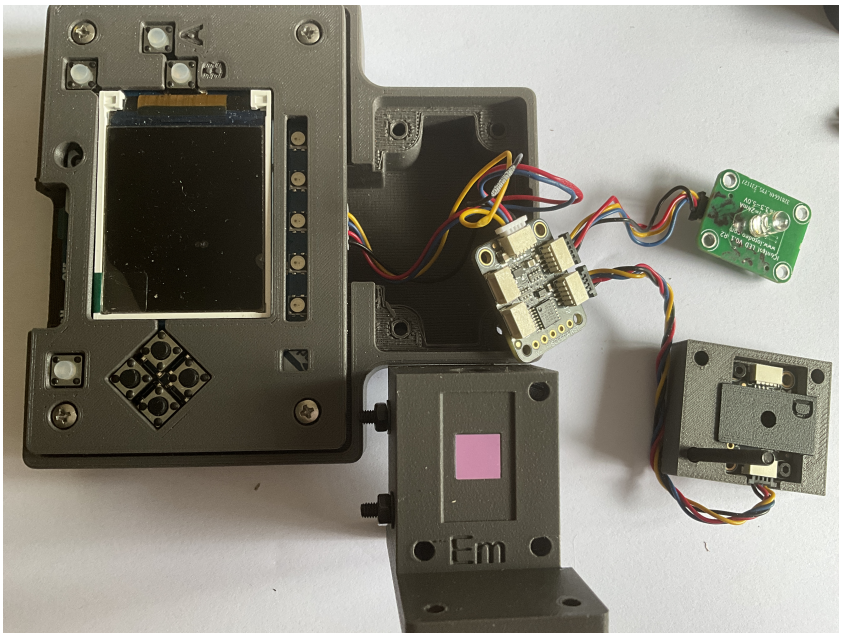

Figure 4.9: Interior view of the assembled instrument showing the PyBadge, PCA9546 I²C multiplexer, modified ioRodeo LED board, TSL2591 sensor and internal wiring.

4.10 STEMMA QT / Qwiic wiring and I²C addresses

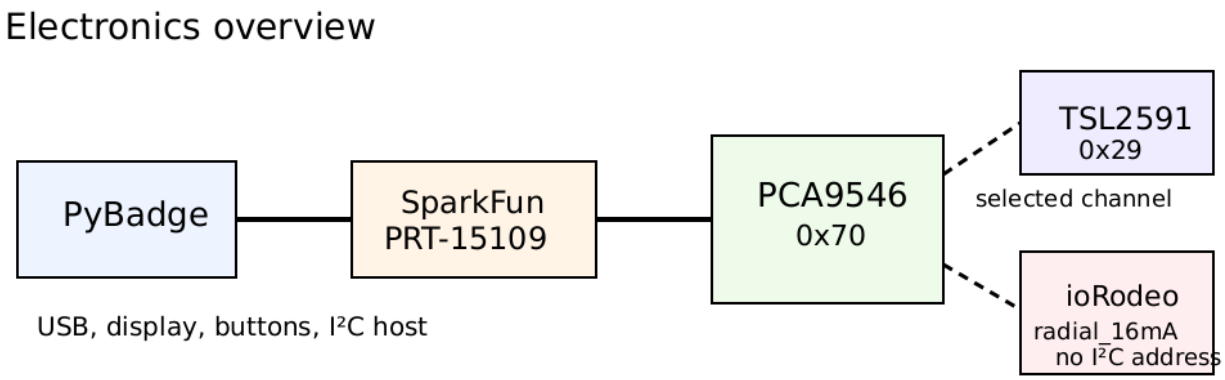

Figure 4.10: Electronics overview. The TSL2591 may be connected to any available downstream PCA9546 channel. The firmware scans the downstream channels for the sensor at address `0x29`; therefore, a fixed downstream channel assignment is not required.

The confirmed address map is:

Device	Address	Expected in I²C scan?
PCA9546 multiplexer	`0x70`	yes, on the upstream bus
TSL2591	`0x29`	yes, on its selected downstream channel
ioRodeo fixed 16 mA LED board	none	no

The intended upstream connection is:

`PyBadge I²C connector → SparkFun PRT-15109 → PCA9546 upstream connector`

The exact downstream PCA9546 channel assignment, cable order, and cable lengths remain to be transcribed from the final instrument. They are deliberately marked `TBD` in the cable map rather than guessed.

When inserting JST-SH connectors:

- hold the plug, not the cable;
- check alignment before applying pressure;
- do not lever the socket from the PCB;
- do not connect or disconnect while the instrument is powered;
- verify that no cable crosses the tube opening or blocks a filter aperture.

4.11 Cable routing, strain relief, and prevention of light leakage

1. Route cables along enclosure walls and printed channels.
2. Keep the optical chamber clear of wire loops.
3. Provide a service loop near each detachable board.
4. Add strain relief close to the PyBadge adapter and optical module, not directly at the JST-SH socket.
5. Avoid sharp folds in Qwiic cables.
6. Keep cables clear of the reset button, power switch, and USB connector.
7. Close unused optical and cable openings with the intended printed cover or opaque material.
8. Inspect the closed enclosure in a dark room while illuminating its exterior from several directions.
9. Pay particular attention to the tube opening, filter pockets, cable pass-throughs, display edge, and the joint between the optical module and base.

The final light shield must be fitted before blank stability and low-signal performance are evaluated.

4.12 Recommended step-by-step assembly sequence

1. Print and inspect all mechanical parts.
2. Test the E4941 tube fit using an empty tube.
3. Test filter-pocket fit using a **8 x 8 x 1 mm** test piece without touching the optical apertures.
4. Install the 485 nm LED on the ioRodeo revision-3 board.
5. Install the Ex470BP-40 excitation filter.
6. Install the E4941 sample-tube holder.
7. Install the Em532BP-40 emission filter.
8. Install the single TSL2591 with its active area facing the filter.
9. Mount the optical module on the instrument base.
10. Mount the PyBadge in its enclosure.
11. Mount the PCA9546 and LED board.
12. Connect the upstream adapter and downstream Qwiic cables according to the final cable map.
13. Route cables and add strain relief.
14. Fit the light shield.
15. Perform the pre-power inspection.
16. Carry out first power-up using USB before adding an optional LiPo battery.
17. Record the actual channel map, cable lengths, and any assembly deviations.

4.13 Complete Measurement Setup

Figure 4.11: Complete measurement setup including the DIY-QuantiFluorONE-dsDNA-Fluorometer, Promega QuantiFluor® ONE dsDNA System, pipettes, PCR tubes and the supporting laboratory equipment used for routine measurements.

4.14 Pre-power inspection

Do not power the instrument until every item below is checked.

- ☐ PyBadge is switched off.
- ☐ USB is disconnected.
- ☐ LiPo battery is disconnected or confirmed undamaged and correctly connected.
- ☐ LED polarity is correct.
- ☐ No LED lead or solder joint contacts the enclosure hardware.
- ☐ Ex470BP-40 is installed on the LED side.
- ☐ Em532BP-40 is installed on the sensor side.
- ☐ TSL2591 active area faces the emission filter.
- ☐ Only one TSL2591 is installed.
- ☐ PCA9546 orientation and upstream connection are correct.
- ☐ All JST-SH plugs are fully seated and correctly aligned.
- ☐ No cable is pinched by the cover.
- ☐ Reset and power controls move freely.
- ☐ USB Micro connector is accessible without forcing the cable.

☐ Tube can be inserted and removed without contacting electronics.
☐ Light shield and enclosure close without pressure on the boards.
☐ No loose screw, wire clipping, or printed debris remains inside.

4.15 First power-up and hardware checks

Use USB power for the first check. Leave the optional LiPo disconnected.

1. Connect a known-good USB Micro data cable.
2. Switch on the PyBadge.
3. Confirm that the display starts normally.
4. Confirm that the PCA9546 is detected at `0x70`.
5. Select the configured downstream channel and confirm that the TSL2591 is detected at `0x29`.
6. Do not expect an I²C address for the fixed-current LED board.
7. Confirm that the 485 nm LED illuminates when the hardware and firmware configuration calls for it.
8. Do not look directly into the LED or the excitation aperture.
9. Place an empty E4941 tube in the holder and close the light shield.
10. Compare the sensor reading with the shield open and closed. A large room-light response indicates incomplete shielding or an uncovered optical path.
11. Insert the intended blank and check that repeated readings are stable enough for subsequent calibration work.
12. Record any channel or cable-map corrections immediately.

Firmware installation, menu operation, and analytical calibration are covered in their own chapters.

4.16 Troubleshooting

Symptom	Likely cause	Workshop check
PyBadge does not start	power switch off, charge-only/failed cable, low battery, short circuit	disconnect external modules; test USB cable and switch position
PCA9546 not found at `0x70`	adapter orientation, loose connector, damaged cable, upstream wiring error	inspect PyBadge-to-PRT-15109-to-PCA9546 chain
TSL2591 not found at `0x29`	wrong channel, loose downstream cable, incorrect orientation, damaged socket	select each documented channel; inspect cable and sensor board
LED board not shown in scan	normal behavior	the fixed 16 mA board has no I²C address
LED remains dark	reversed LED, poor solder joint, no supply, damaged LED or driver board	power off; verify polarity and continuity before replacement
Sensor saturates	light shield open, emission filter missing, filter displaced, direct LED path	close shield and inspect filter sequence and geometry
Signal is unexpectedly low	LED/filter misalignment, sensor faces away, dirty filter, tube not seated	inspect optical axis, active area, tube stop, and optical surfaces

Symptom	Likely cause	Workshop check
Blank changes with room lighting	light leak at tube opening, cable slot, enclosure seam, or display edge	dark-room flashlight inspection; refit shield and covers
Tube jams	support residue, undersized bore, damaged tube	remove tube; clean or reprint holder; do not force it
Readings change after handling	loose filter, sensor board movement, cable strain	inspect retention and strain relief; repeat mechanical check

4.17 Cleaning, maintenance, and replacement of optical components

- Switch off and disconnect USB and LiPo before opening the instrument.
- Remove the sample tube immediately after use.
- Wipe spilled reagent before it reaches filter pockets or PCB surfaces.
- Handle filters by the edges using clean gloves or suitable tweezers.
- Use clean optical tissue or a blower intended for optical work.
- Do not use abrasive paper, cotton fibers, or aggressive solvent on filter coatings.
- Check compatibility before applying alcohol or another solvent to printed parts.
- Do not scrape the active TSL2591 area.
- Replace cracked, delaminated, deeply scratched, or chemically attacked filters.
- Recheck blank response and recalibrate after replacing the LED, either optical filter, the sample holder, or the TSL2591.
- Inspect Qwiic sockets and cables periodically for looseness or strain damage.
- Inspect the LiPo for swelling, puncture, unusual heat, or damaged leads. Remove a suspect battery from service.

4.18 Safety notes

- Do not stare directly into the 485 nm LED, especially when the excitation filter or light shield is removed.
- Disconnect power before soldering, rewiring, installing filters, or moving boards.
- Prevent metal tools and loose screws from shorting the PyBadge or LED board.
- Observe LiPo polarity. Do not use swollen, damaged, wet, or overheated batteries.
- Charge the battery only through the intended PyBadge charging circuit and a suitable USB supply.
- Keep liquids away from the open electronics.
- Follow the safety data sheets and laboratory rules for QuantiFluor ONE reagent, DNA standards, cleaning agents, and samples.
- Dispose of reagents, samples, tubes, electronics, and batteries according to local rules.

4.19 Reproducibility files and references

This package includes:

- PyBadge enclosure STEP files;
- CAD screenshots of the enclosure and optical module;
- ioRodeo `radial_16mA` revision-3 KiCad, Gerber, BOM, CPL;
- Adafruit PyBadge EagleCAD schematic and board files;
- upstream license files and source notes;
- the current assembled-instrument photograph;
- BOM, wiring, I²C, figure, and open-item manifests;
- STEP-Files.

The third-party source locations and licenses are recorded in `THIRD_PARTY_NOTICES.md` and the `SOURCE.md` files beside the upstream hardware data.

5 Software Installation

Current supported software set: PyBadge UF2 bootloader 3.15.0, Adafruit CircuitPython 9.1.1, and the unchanged `QuantiFluorONE_Firmware_1.0.0-rc2-stable.zip` release.

Repository policy: one instrument, one supported QuantiFluorONE firmware release, and one documented installation procedure.

5.1 Purpose and scope

This chapter describes the workshop procedure for preparing an Adafruit PyBadge and installing the current QuantiFluorONE firmware. It covers:

- backing up an existing `CIRCUITPY` drive;
- distinguishing normal restart, CircuitPython safe mode, and UF2 bootloader mode;
- checking or updating the PyBadge bootloader;
- installing CircuitPython 9.1.1;
- performing a clean installation of the complete stable RC2 release;
- carrying out the first startup and basic file-system checks.

The physical reset button is mounted on the **underside of the PyBadge**. It is reached from below through the dedicated reset access opening in the enclosure.

5.2 Software files in the repository

All installation components are stored under one top-level directory:

```
software/
  README.md
  bootloader/
      3.15.0/
          update-bootloader-arcade_pybadge-v3.15.0.uf2
          bootloader-arcade_pybadge-v3.15.0.bin
          README.md
  circuitpython/
      9.1.1/
          adafruit-circuitpython-pybadge-en_US-9.1.1.uf2
          README.md
  quantifluorone_firmware/
      QuantiFluorONE_Firmware_1.0.0-rc2-stable.zip
      README.md
      SHA256SUMS.txt
```

Component: PyBadge bootloader updater
Repository file:
`software/bootloader/3.15.0/`
`update-bootloader-arcade_pybadge-v3.15.0.uf2`
When required: Only when the bootloader must be updated

Component: Raw bootloader image
Repository file:
`software/bootloader/3.15.0/`
`bootloader-arcade_pybadge-v3.15.0.bin`
When required: Archive and technical recovery only

Component: CircuitPython
Repository file:
`software/circuitpython/9.1.1/`
`adafruit-circuitpython-pybadge-en_US-9.1.1.uf2`
When required: After a bootloader update or for a clean CircuitPython installation

---

Component: Current device firmware
Repository file:
`software/quantifluorone_firmware/`
`QuantiFluorONE_Firmware_1.0.0-rc2-stable.zip`
When required: Clean QuantiFluorONE installation

Use a computer with a USB port and a known-good USB Micro **data cable**. A charge-only cable cannot expose the PyBadge drives.

Use a short **non-conductive** tool to reach the reset switch through the underside opening. The eraser end of a pencil or a suitable plastic rod is preferable to a metal tool.

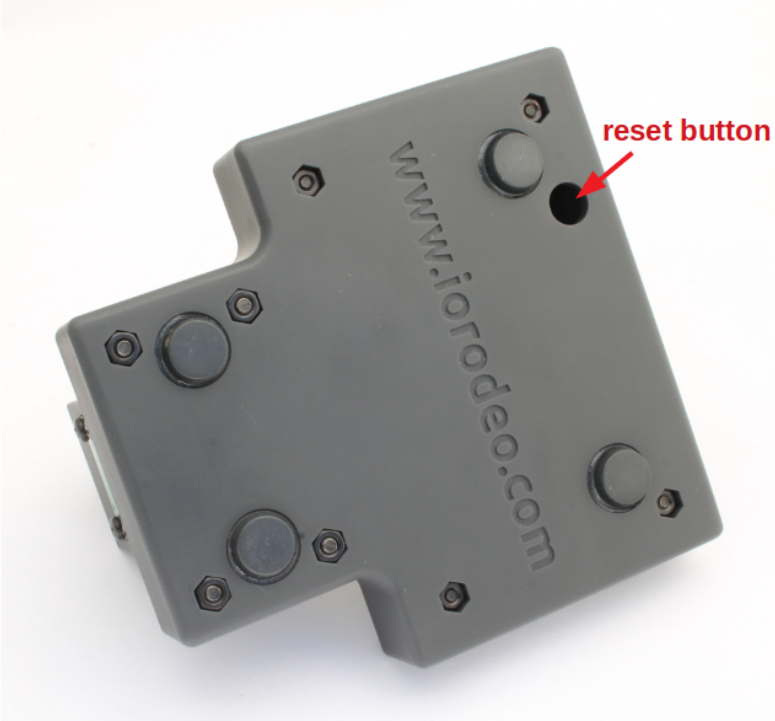

Figure 5.1: The reset switch is mounted on the underside of the PyBadge and is reached from below through the enclosure opening.

5.3 One supported QuantiFluorONE firmware

The current supported device firmware is the unchanged archive:

```
software/quantifluorone_firmware/
QuantiFluorONE_Firmware_1.0.0-rc2-stable.zip
```

Do not shorten, clean, rename, rebuild, or replace this archive with a reduced derivative. The ZIP file is the preserved software state that was tested on the instrument.

Earlier development versions remain traceable through Git history, tags, or GitHub releases, but they are not presented in the main repository as competing installation choices.

> **Important:** "stable RC2" describes the tested firmware package and operating workflow. The documented multipoint calibration and analytical workflow have been evaluated using the validation procedure described in Chapter 10.

5.4 Back up the current device

> **Back up the complete `CIRCUITPY` drive before continuing.** Deleting its contents also removes measurement logs, calibration files, configuration files, saved blank and state files, and other user-generated data stored on the device.

1. Connect the instrument using a USB data cable.
2. Switch the PyBadge on.
3. Open the `CIRCUITPY` drive if it appears.

4. Copy its complete contents to a dated backup folder on the computer.
5. Confirm that the backup includes all CSV and JSON files that must be retained.
6. Safely eject `CIRCUITPY` before resetting, disconnecting, or power-cycling the instrument.

Do not copy old configuration, state, calibration, or log files back to the device automatically after installation. Restore only files that are known to be compatible with the current firmware.

5.5 Reset timing: three different results

Reset action	Result	Drive	Main use
One press	normal restart	`CIRCUITPY`	restart the application or leave safe mode
Slow double-click	CircuitPython safe mode	`CIRCUITPY`	copy, replace, download, or delete firmware, CSV, and JSON files
Fast double-click	UF2 bootloader mode	`PYBADGEBOOT` or `BADGEBOOT`	install a bootloader-updater or CircuitPython UF2 file

5.5.1 Slow double-click: required for file maintenance

Use the **slow double-click** before modifying files on `CIRCUITPY`:

1. Press the underside reset button once through the access opening.
2. Allow the startup sequence to begin.
3. Press Reset again during the approximately one-second safe-mode window.

Safe mode prevents the QuantiFluorONE application from running while files are changed. Automatic reload is disabled, making this the preferred state for copying firmware, replacing JSON files, downloading CSV logs, and deleting old files.

5.5.2 Fast double-click: UF2 bootloader mode

Press the underside reset button twice in quick succession. The PyBadge display changes to the UF2 bootloader screen, and the computer mounts `PYBADGEBOOT` or `BADGEBOOT`.

Figure 5.2: PyBadge UF2 bootloader screen.

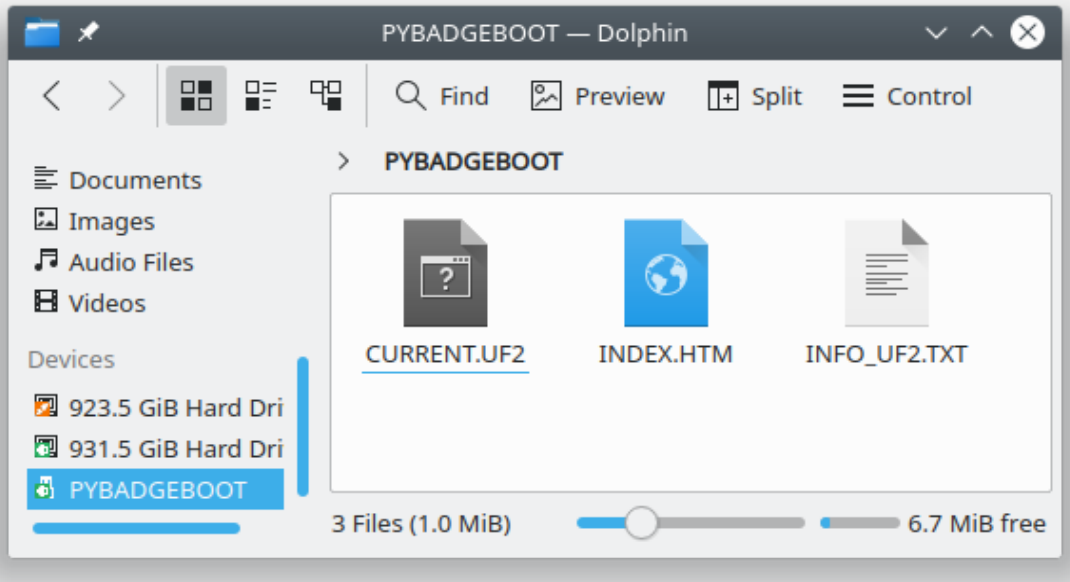

Figure 5.3: Typical contents of the PyBadge UF2 boot drive.

If `PYBADGEBOOT` appears when `CIRCUITPY` was intended, press Reset once to restart normally and repeat with slower timing.

5.6 Check or update the UF2 bootloader

A bootloader update is not required for every firmware installation.

1. Enter UF2 bootloader mode with a fast double-click.
2. Open `INFO_UF2.TXT` on `PYBADGEBOOT` or `BADGEBOOT`.
3. Read the installed bootloader version.

To install the supplied version 3.15.0 updater:

1. Copy `update-bootloader-arcade_pybadge-v3.15.0.uf2` to the UF2 boot drive.
2. Wait while the updater runs and the boot drive reappears.
3. Open `INFO_UF2.TXT` again and verify version 3.15.0.
4. Install CircuitPython 9.1.1 as described in the next section.

   Do not copy `bootloader-arcade_pybadge-v3.15.0.bin` to `PYBADGEBOOT` or `BADGEBOOT`. The BIN file is retained only for archival and technical recovery purposes.

5.7 Install CircuitPython 9.1.1

1. Confirm that the current `CIRCUITPY` contents have been backed up.
2. Enter UF2 bootloader mode with a **fast double-click**.
3. Copy `adafruit-circuitpython-pybadge-en_US-9.1.1.uf2` to `PYBADGEBOOT` or `BADGEBOOT`.
4. Wait for the board to restart automatically.
5. Confirm that the UF2 boot drive disappears and a new drive named `CIRCUITPY` appears.
6. Safely eject `CIRCUITPY` before disconnecting or restarting the instrument.

The stable RC2 archive records the tested runtime in `boot_out.txt` as:

```
Adafruit CircuitPython 9.1.1 on 2024-07-22
Adafruit Pybadge with samd51j19
Board ID: pybadge
```

5.8 Clean installation of QuantiFluorONE stable RC2

This is the current recommended workshop procedure.

5.8.1 Prepare the firmware folder

1. Locate:

   ```
   software/quantifluorone_firmware/
   QuantiFluorONE_Firmware_1.0.0-rc2-stable.zip
   ```

2. Extract the ZIP file to a normal folder on the computer.
3. Open the extracted outer folder named:

   ```
   QuantiFluorONE_Firmware_1.0.0-rc2-stable
   ```

4. Leave the original ZIP file unchanged in the repository.

5.8.2 Empty `CIRCUITPY`

1. Confirm once more that the complete previous drive has been backed up.
2. Enter safe mode with the **slow double-click**.
3. Open the root of `CIRCUITPY`.
4. Select all existing files and folders on `CIRCUITPY`.
5. Delete the selected contents.
6. Confirm that no old firmware, library, configuration, state, calibration, or log files remain.

5.8.3 Copy the complete stable release

1. In the extracted stable-release folder, select **all files and folders inside** `QuantiFluorONE_Firmware_1.0.0-rc2-stable`.
2. Copy the complete selected contents to the **root of the empty `CIRCUITPY` drive**.
3. Do **not** copy the enclosing `QuantiFluorONE_Firmware_1.0.0-rc2-stable` folder itself. Files such as `boot.py` and `code.py` must be located directly in the root of `CIRCUITPY`.
4. Wait until copying is completely finished.
5. Compare the top-level files and folders with the extracted release folder.
6. Safely eject `CIRCUITPY`.
7. Press Reset once or power-cycle the device for a normal restart.

This full-copy procedure installs one complete, internally consistent release and avoids mixing files from different firmware generations.

5.9 First startup and hardware check

After the normal restart:

1. Confirm that the splash screen appears.
2. Confirm the display banner:

   ```
   QuantiFluorONE v1.0 RC2
   ```

3. Confirm that no startup error screen is shown.
4. Open the SELECT menu.
5. Choose `LOAD MP CAL` only when the supplied or a compatible `quantifluorone_multipoint.json` is present.
6. Measure a fresh reagent blank with START.
7. Measure a test sample with A.

The display may show `cal=none c=uncalibrated` until a calibration is loaded or created. After loading a compatible calibration, verify that the calibration information displayed by the instrument corresponds to the calibration intended for the current measurement series.

5.10 CSV and JSON write behaviour

The included `boot.py` provides two different file-system modes:

- with a USB data host connected, `CIRCUITPY` remains writable by the computer and read-only to the firmware;
- with a charger, power bank, or approved LiPo supply, the firmware can save blank state, calibration state, and CSV rows.

For firmware-generated writes:

1. safely eject and disconnect the USB data host;
2. power the PyBadge from a USB charger, power bank, or the approved LiPo supply;
3. restart the instrument normally;
4. perform the measurement or save operation.

Use the **slow double-click** whenever returning to computer-based file maintenance.

5.11 Updating or reinstalling the same supported release

Use the same clean procedure whenever a complete reinstall is needed:

1. back up `CIRCUITPY`;
2. enter safe mode with the slow double-click;
3. delete the complete contents of `CIRCUITPY`;
4. extract the unchanged stable RC2 archive;
5. copy all contents inside the extracted release folder to the root of `CIRCUITPY`;
6. safely eject and restart;
7. verify the RC2 display banner.

Do not merge a new installation with residual files from an older device state.

5.12 Troubleshooting

Symptom	Likely cause or corrective action
No USB drive appears	check the power switch, USB port, and use a known data-capable cable
`PYBADGEBOOT` appears instead of `CIRCUITPY`	reset presses were too fast; press once to restart, then use the slow double-click if file maintenance is required
`CIRCUITPY` appears but files cannot be changed reliably	enter safe mode with the slow double-click before copying or deleting files
Device remains in safe mode	safely eject, then press Reset once or power-cycle
PyBadge repeatedly reloads while files are copied	use safe mode so that the application is not running during file maintenance
Firmware starts with missing-module or missing-file errors	repeat the clean installation and confirm that the complete release contents were copied to the root of `CIRCUITPY`

Symptom	Likely cause or corrective action
`boot.py` or `code.py` is not visible in the root of `CIRCUITPY`	the enclosing release folder was copied instead of its contents; empty the drive and repeat the installation
Firmware starts but does not log CSV while connected to a computer	expected project file-system behaviour; use a charger, power bank, or LiPo supply for firmware-side writes
Old measurement or calibration state reappears	residual files were retained or restored; repeat the clean installation and restore only confirmed compatible files

The following screen indicates CircuitPython automatic reload or a soft reboot, not UF2 bootloader mode:

Figure 5.4: CircuitPython soft-reboot and automatic-reload screen.

5.13 References

- IO Rodeo. *Installing/upgrading CircuitPython and the bootloader.* https://blog.iorodeo.com/installing-upgrading-circuitpython-or-the-bootloader/
- IO Rodeo. *Installing/upgrading the firmware.* https://blog.iorodeo.com/installing-upgrading-the-firmware/
- Adafruit. *Installing CircuitPython — Adafruit PyBadge and PyBadge LC.* https://learn.adafruit.com/adafruit-pybadge/installing-circuitpython
- Adafruit. *Update the PyBadge Bootloader.* https://learn.adafruit.com/adafruit-pybadge/updating-the-bootloader

- Adafruit. *Troubleshooting — Safe Mode.* https://learn.adafruit.com/adafruit-pybadge/troubleshooting
- CircuitPython.org. *PyBadge downloads.* https://circuitpython.org/board/pybadge/

This chapter adapts the general IO Rodeo and Adafruit procedures to the current QuantiFluorONE hardware, enclosure, software versions, and the tested clean-installation workflow used for this instrument.

6 Getting Started: Basic Operation and Menu Navigation

Firmware basis: QuantiFluorONE `QF1-1.0.0-rc2`, display version `v1.0 RC2`.

6.1 Front-panel controls

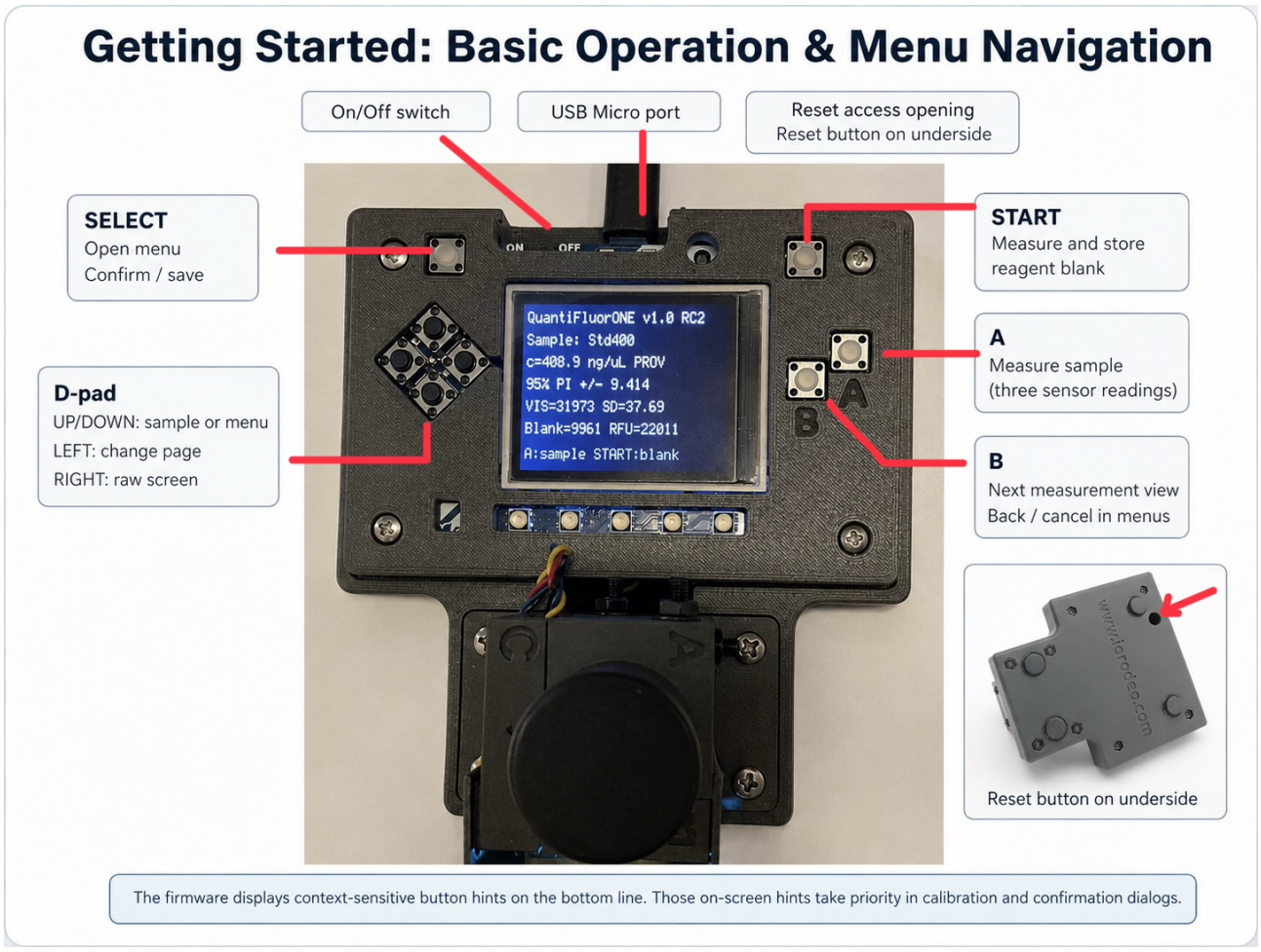

Figure 6.1: QuantiFluorONE front-panel controls for firmware QF1-1.0.0-rc2.

The firmware defines the controls as follows:

Physical control	Firmware name	Main measurement-screen function	Menu or dialog function
Upper-left button	`SELECT`	Open the SELECT menu	Open, confirm, load, or save the highlighted action
Upper-right button	`START`	Measure and store the reagent blank	Measure a calibration blank or the next blank in the ten-blank study
A button	`A`	Measure the selected sample; one cycle contains three sensor readings	Measure the selected high standard during two-point calibration
B button	`B`	Cycle Live → Details → RAW → Live	Back or cancel
D-pad UP/DOWN	`UP/DOWN`	Select the sample ID	Select a menu item or standard concentration
D-pad RIGHT	`RIGHT`	Open or leave the RAW screen	Not used for menu selection
D-pad LEFT	`LEFT`	Change RAW page	Change calibration-result or calibration-status page
On/Off switch	power	Turn the instrument on or off	—
Reset access opening	reset	Restart the PyBadge	Used for normal reset, slow-double-click safe mode, or fast-double-click UF2 bootloader mode
USB Micro port	USB	Power, charging, programming, and file access	—

The last line of the display shows context-sensitive button hints. These hints take priority whenever a calibration or confirmation dialog is open.

6.2 Quick 2-PT calibration and measurement

- Take standard and fluorescent dye out of the freezer/fridge and warm up to room temperature.
- Please switch the instrument on and wait for 20 Minutes before you start the first measurements.
- During the warm up prepare the Blank and the 400 ug/uL-Standard for the 2-PT calibration in 0.5 mL tubes.
- Press SELECT button for 2 time to start the 2-PT calibration of 9the instrument.
- Put the Blank-sample (200 uL Promega QuantiFluor® ONE dsDNA Dye) into the cuvette holder.
- Press START button to measure the blank.
- Remove the blank and store it in the dark.
- Put the 400 ug/uL-standard (200 uL Promega QuantiFluor® ONE dsDNA Dye + 1 uL QuantiFluor® ONE Lambda DNA, mix well) into the cuvette holder.
- Press SELECT button
- Press A button to measure the standard.

- Press SELECT button to save the 2-PT calibration data.
- The expected values are; slope ~114.5 ± 0.5 ; Blank raw ~ 6500 - 8500 ; 400 ng/uL ~ 45800 ± 500 RFU
- Press B button to go back to the measurement.
- Select “sample01” with the D-Pad up/down keys.
- Now you should measure your samples 200 uL Promega QuantiFluor® ONE dsDNA Dye + 1 uL sample. Mix well. Do not forget to write a protocol of your measurements.

For our Amlicon with ~ 726 bp a rule of thumb -> 48 ng/µL ~ 100 fmol. The maximum error with slope of 114.7 ± 0.5 and Blank raw of 6000-9000 ~ ±25 fmol.

6.3 Preparing and documenting the sample list

Before starting a measurement series, the sample identifiers can be entered in `quantifluorone_config.json`. This is recommended because the selected sample ID is written to the measurement CSV file and allows each measurement to be assigned directly to the corresponding laboratory sample.

Editing the sample list is optional. If the list is not updated, the ebhxact order in which the physical samples are measured must be recorded separately in the laboratory notebook. Without such a record, measurements in the CSV file may no longer be assignable reliably to the original samples after the measurement series.

6.3.1 Editing the sample list

1. Connect the switched-on fluorometer to the computer using a USB data cable.
2. Enter CircuitPython safe mode using the **slow double-click** of the Reset button:
   - press Reset once;
   - allow the restart sequence to begin; and
   - press Reset a second time during the approximately one-second startup window.
3. Confirm that the `CIRCUITPY` drive is available on the computer.
4. Open `quantifluorone_config.json` with a plain-text editor.

   Suitable examples are:

   - Windows Notepad;
   - Notepad++;
   - Visual Studio Code; or
   - another editor that saves plain-text files without changing the JSON structure.

   Do not edit the file with a word processor such as Microsoft Word.
5. Locate the `samples` section and enter the required sample identifiers, for example:

```
"samples": {
  "list": ["Blank", "Std400", "QC1", "Sample01", "Sample02"],
  "selected_index": 0
}
```

Use short, unique identifiers because the display width is limited. The complete identifier is retained in the measurement data even when only part of it is visible on the instrument display.

6. Save `quantifluorone_config.json` without changing its filename or file type.
7. Safely eject the `CIRCUITPY` drive from the computer.
8. Disconnect the USB data connection.
9. Connect the fluorometer to its normal standalone power source, such as a power bank, and restart it normally.
10. Use **UP/DOWN** on the measurement screen to select the required sample ID before measuring the corresponding sample with **A**.

6.3.2 If the sample list is not edited

Updating `quantifluorone_config.json` is not mandatory. The instrument can also be used with the existing sample list.

In this case, record the physical samples and their measurement order explicitly in the laboratory notebook, for example:

```
Measurement 1  = Sample A17
Measurement 2  = Sample A18
Measurement 3  = Sample B03
Measurement 4  = Sample B04
```

Important: The CSV file records the sample identifier selected on the instrument. If this identifier does not correspond to the actual sample and no separate measurement-order record is kept, the analytical result may not be assignable reliably to the original sample afterwards.

Whenever practical, prepare the sample list before the measurement series and keep the displayed sample order consistent with the physical order of the prepared tubes.

6.4 Normal measurement workflow

1. Switch the instrument on and wait for `QuantiFluorONE v1.0 RC2`.
2. Use D-pad UP or DOWN to select the required sample ID.
3. Insert a fresh reagent blank and close the light shield.
4. Press START. The instrument performs three TSL2591 readings and stores the blank mean and standard deviation.
5. Replace the blank with the prepared sample and close the light shield.
6. Press A. The instrument performs one sample cycle of three sensor readings.
7. Read concentration, prediction interval, VIS, standard deviation, blank, and RFU from the live screen.
8. Press B for the details screen or RIGHT for raw readings.

6.5 SELECT menu

Press SELECT from the live screen to open:

```
SELECT MENU
>2-PT CAL
 10-BLANK L/Q
 LOAD MP CAL
 CAL STATUS
 CLEAR CAL
```

Use UP/DOWN to move the selection, SELECT to open or confirm, and B to return or cancel.

6.6 Display views

6.6.1 Live

The live screen displays the sample ID, concentration or calibration status, prediction interval or LOD/LOQ, VIS mean and SD, blank, and RFU.

6.6.2 Details

Press B from Live. The details screen shows FULL, IR, VIS, RFU, concentration, sensor channel, gain, and integration time.

6.6.3 RAW

Press B again, or press RIGHT from Live/Details. LEFT changes between the two raw-data pages. RIGHT returns to Live.

6.7 Multipoint calibration import

1. Place the compatible `quantifluorone_multipoint.json` file in the root of `CIRCUITPY`.
2. Disconnect the USB data host and power the device from a charger or power bank so the firmware can write its state files.
3. Press SELECT.
4. Choose `LOAD MP CAL`.
5. Confirm with SELECT.
6. Measure a fresh reagent blank with START before measuring samples.

The multipoint calibration used for quantitative measurements must correspond to the validated calibration workflow described in this repository. Verify the calibration identity, working range, concentration basis, and status before measuring samples.

6.8 Managing CSV and JSON files: use the slow double-click

> **Important — slow double-click required:** Before copying, replacing, deleting, or downloading CSV, JSON, or firmware files, enter CircuitPython **safe mode** with a slow double-click of the Reset button. Press Reset once, then press it again during the approximately one-second startup window. The display reports that code has stopped and will not run the saved application.

In safe mode:

- `CIRCUITPY` remains available to the computer;
- the QuantiFluorONE application does not run;
- automatic reload is disabled;
- files can be copied, downloaded, replaced, or deleted without the firmware actively using them.

A **fast** double-click is different: it opens `PYBADGEBOOT` or `BADGEBOOT` and is used only for UF2 installation.

6.9 Files produced or used by RC2

File	Purpose
`quantifluorone_log_v100rc2.csv`	Measurement log
`quantifluorone_state.json`	Blank and interface state
`quantifluorone_calibration.json`	Active saved calibration
`quantifluorone_multipoint.json`	Multipoint model imported through `LOAD MP CAL`
`quantifluorone_config.json`	Hardware, measurement, sample, assay, and calibration settings

Always back up required data and safely eject the drive before resetting, switching off, or disconnecting USB.

7 Calibration Suite and Data Transfer

7.1 Purpose

The repository includes the current browser-based Calibration Suite as a local web app. This chapter covers the calibration workflow and the safe transfer of compatible calibration files to the QuantiFluorONE firmware.

7.2 Repository location

The current files are stored in:

- `tools/calibration_suite/Open_Fluorometer_Calibration_Suite_v7.3.html`
- `tools/calibration_suite/Open_Fluorometer_Calibration_Suite_v7.3.txt`

7.3 Basic workflow

1. Open the HTML file in a current desktop browser.
2. Load or paste the calibration data required by the app.
3. Review the plotted data and the resulting calibration output.
4. Record the calibration parameters required for the fluorometer workflow.
5. Store calibration exports together with the experimental metadata whenever possible.

7.4 Recommended repository practice

- keep the web app under version control;
- record the active calibration-suite version in documentation and workshop notes;
- store example calibration data separately from the app itself;
- treat the web app as a reproducibility tool, not merely as a convenience file.

7.5 Calibration-file transfer

1. Export and retain the complete calibration report and source data.
2. Transfer only a file that is explicitly compatible with QuantiFluorONE `QF1-1.0.0-rc2`.
3. Enter CircuitPython safe mode with the slow double-click before replacing JSON files on `CIRCUITPY`.
4. Back up the existing JSON file before replacement.
5. Safely eject `CIRCUITPY`, restart normally, and confirm the calibration status on the instrument.

8 Measurement Protocol

This short protocol is intended for participants with previous laboratory experience. All reagents required for the dsDNA assay are supplied with the Promega QuantiFluor® ONE dsDNA System. No reagent preparation is required.

8.1 Materials and preparation

8.1.1 Laboratory materials

- Powder-free latex gloves
- P2 or P10 pipette with sterile, DNA-free tips
- 200 µL pipette with sterile, DNA-free tips
- 1,000 µL pipette with sterile, DNA-free tips, if required for batch handling
- Promega QuantiFluor® ONE dsDNA System, 100 assays
- Thin-walled Promega E4941 0.5 mL PCR tubes
- Tube rack
- Timer
- Drawer or another dark place for light-protected incubation
- Notebook and pencil

8.1.2 Equipment and digital access

- DIY-QuantiFluorONE-dsDNA-Fluorometer
- Smart device with access to the project GitHub repository and the Calibration Suite

8.2 Clean working conditions

Work on a clean bench and wear fresh gloves. All pipette tips, PCR tubes, and other materials that come into contact with samples, standards, or reagents must be sterile and DNA-free.

The fluorometer, smart device, notebook, and pencil do not need to be sterile, but they must be kept clean and away from open tubes. Replace gloves immediately after possible contact with DNA-containing material.

8.3 Assay setup

1. Label one PCR tube for the blank and additional tubes for standards, quality-control samples, and unknown samples.
2. Mix the QuantiFluor® ONE reagent and the DNA solutions gently before use. Avoid foaming.
3. Pipette 200 µL of QuantiFluor® ONE reagent into each tube.
4. Add 1 µL of blank solution, standard, quality-control sample, or unknown sample with a P2 or P10 pipette.
5. Close each tube and mix gently without creating bubbles.
6. Keep all tubes protected from direct light.

Use a fresh pipette tip for every transfer.

8.4 Incubation

Place the prepared tubes in the rack and incubate them in a closed drawer or another dark place. Use the same incubation time for the blank, standards, quality-control samples, and unknown samples.

Incubate the prepared tubes for 5 minutes at room temperature, protected from light, in accordance with the Promega QuantiFluor® ONE dsDNA System instructions. Start the timer immediately after preparation of the last tube or prepare larger sample sets in clearly timed batches.

8.5 Blank, calibration, and sample measurement

1. Switch on the DIY-QuantiFluorONE-dsDNA-Fluorometer.
2. Open the Calibration Suite on the smart device when calibration data must be loaded, checked, or evaluated.
3. Select or load the calibration intended for the current measurement series.
4. Insert the blank tube and perform the blank measurement.
5. Measure the calibration standards or quality-control samples when required.
6. Measure the unknown samples one after another.
7. Insert every tube in the same way and ensure that it is fully seated in the holder.
8. Record the displayed concentration and the accompanying quality information.

Do not measure tubes containing visible bubbles, droplets on the outside, or fingerprints in the optical measurement area. Clean or replace the tube before repeating the measurement.

8.6 Recording and checking results

Record at least:

- Sample identification
- Date and time
- Calibration used
- Displayed dsDNA concentration
- Measurement error or uncertainty shown by the instrument
- LOD and LOQ shown by the instrument
- Dilution factor, if the original sample was diluted
- Relevant observations, such as bubbles or repeated measurements

Results below the LOD must not be reported as quantified concentrations. Results between the LOD and LOQ may be reported only as detected but not reliably quantified. Samples above the valid calibration range must be diluted and measured again.

When replicate measurements are performed, report the mean value and note any clearly inconsistent result.

8.7 Cleaning and completion

Dispose of used tubes and pipette tips according to the local laboratory rules. Wipe the outside of the fluorometer and clean the working area. Do not allow liquid or cleaning agent to enter the fluorometer.

Store the QuantiFluor® ONE kit components according to the manufacturer's instructions and ensure that all required measurement and calibration data have been saved before completing the measurement session.

8.8 Quick reference

1. Put on fresh gloves and label the tubes.
2. Add 200 µL reagent to each tube.
3. Add 1 µL blank, standard, control, or sample.
4. Mix gently and incubate protected from light.
5. Blank the fluorometer and load the correct calibration.
6. Measure standards, controls, and samples.
7. Record concentration, error, LOD, LOQ, and any dilution.

9 DIY-QuantiFluorONE-dsDNA-Fluorometer – Calibration, Results, and Quality Control

Scope: This chapter provides the concise operational workflow for routine laboratory decisions. The theoretical and statistical basis of calibration, prediction intervals, LOD, and LOQ is described in Chapter 3.

9.1 Calibration routes

Use one of the following calibration routes before measuring unknown samples.

Calibration route	Preparation and evaluation	Instrument action	Appropriate use
On-instrument two-point calibration	Blank and one known standard	`SELECT → 2-PT CAL`	Rapid routine measurements within a limited working range
Multipoint calibration	Evaluate calibration CSV data in the Open Fluorometer Calibration Suite v7.3 Stable	Copy the accepted firmware-compatible JSON to `CIRCUITPY`, then use `SELECT → LOAD MP CAL`	Measurements requiring a defined calibration range, prediction error, LOD, LOQ, and documented model checks
Stored calibration	Review the stored calibration identity and status	`SELECT → CAL STATUS`	Continued work under unchanged and verified conditions
No valid calibration	Raw-signal inspection only	Do not report a concentration	Instrument checks and troubleshooting

A calibration that loads successfully is not automatically suitable for the current samples.

9.2 Use the Calibration Suite

The Calibration Suite is the primary tool for creating, evaluating, documenting, and exporting a multipoint calibration.

9.2.1 Prepare the calibration dataset

Use standards prepared with the same:

- QuantiFluor® ONE dsDNA chemistry;
- sample and reagent volumes;
- tube type;

- incubation conditions;
- instrument configuration;
- gain and integration settings; and
- concentration basis intended for the unknown samples.

Import the calibration CSV into the Calibration Suite and confirm that the analytical-signal and reference-value columns were identified correctly.

For the standard assay described in Chapter 8, each tube contains:

- 1 µL standard or sample; and
- 200 µL QuantiFluor® ONE reagent.

The total assay volume is 201 µL.

9.2.2 Evaluate the calibration

In the Calibration Suite:

1. Enter a unique calibration ID and the calibration date.
2. Confirm the reference-value basis and concentration unit.
3. Calculate the intended calibration model.
4. Inspect the calibration curve and residual plot.
5. Review replicate precision, recovery, outlier flags, and range coverage.
6. Review the slope, intercept, residual error, LOD, LOQ, and prediction statistics.
7. Confirm that no systematic trend, unexplained outlier, or unsuitable range remains.
8. Save or print the calibration report.

Do not accept a calibration from the coefficient of determination alone. A high (R^2) does not compensate for degraded standards, poor pipetting, an unsuitable range, or systematic residuals.

The current QF1-1.0.0-rc2 firmware imports an OLS multipoint calibration. Do not load another model type unless the firmware release documentation explicitly confirms compatibility.

9.2.3 Export the operational calibration

After the model has been accepted:

1. Export the accepted multipoint calibration from the Calibration Suite as `quantifluorone_multipoint.json`.
2. Retain the original CSV, Calibration Suite report, and exported JSON together.
3. Do not edit individual coefficients manually after export.

The Calibration Suite report is the complete calibration record. The JSON file is the operational copy used by the fluorometer.

9.3 Load and verify a multipoint calibration

To transfer an accepted multipoint calibration:

1. Connect the switched-on instrument to a computer with a USB data cable.
2. Make `CIRCUITPY` writable from the computer as described in Chapter 7.
3. Copy `quantifluorone_multipoint.json` to the root of `CIRCUITPY`.
4. Replace the previous file only when the new calibration has been checked and archived.

5. Safely eject and disconnect `CIRCUITPY`.
6. Power the instrument from its normal standalone power source.
7. Open `SELECT → LOAD MP CAL`.
8. Confirm the import.
9. Open `SELECT → CAL STATUS`.

Check at least:

- calibration ID;
- calibration type;
- calibration status;
- calibration range;
- concentration basis and unit;
- LOD and LOQ; and
- availability of the prediction error.

Measure and store a fresh reagent blank after loading a new calibration.

> Loading a new JSON does not replace the need for a current blank and passing QC samples.

9.4 Check calibration validity before each batch

Use a stored or imported calibration only when all of the following are true:

- the calibration status is acceptable for the intended work;
- the calibration ID and date are documented;
- the assay, tube type, volumes, and concentration unit match;
- the instrument hardware and optical configuration are unchanged;
- the gain, integration time, and signal definition match;
- the standards and samples use the same concentration basis;
- the expected sample concentrations are covered by the calibration range;
- the calibration slope is positive;
- required LOD, LOQ, range, and error fields are available; and
- the laboratory-defined validity period has not expired.

Reject or replace the calibration when:

- the wrong file, assay, unit, or concentration basis was selected;
- required fields are missing;
- the expected samples fall outside the validated range;
- the instrument, optical holder, LED, filter, or sensor configuration has changed;
- reagent or standards are known to be degraded; or
- QC performance no longer supports the calibration.

9.5 Measurement-batch sequence

Use the following sequence for each batch:

1. Confirm the active calibration under `CAL STATUS`.
2. Measure a fresh reagent blank.
3. Measure at least one independent QC sample.
4. Measure the unknown samples.

5. Repeat QC after a long interruption, suspected drift, or instrument change.
6. Measure a final QC sample when required by the laboratory procedure.
7. Review the complete batch before accepting results.

For a broad calibration range, preferably use low-, mid-, and high-level QC samples. QC material should be independent of the standards used to create the calibration whenever suitable material is available.

9.6 Evaluate replicate measurements

9.6.1 Three readings within one measurement cycle

The fluorometer uses three TSL2591 readings to calculate one measurement-cycle mean and standard deviation.

These are technical sensor repeats. They are not three independently prepared assays.

Check for:

- an unusually large standard deviation;
- a progressive increase or decrease;
- bubbles in the optical path;
- incomplete mixing;
- incorrect tube seating or orientation;
- fingerprints, droplets, or contamination on the tube; and
- movement or stray-light entry during measurement.

When a visible technical cause is present:

1. correct the problem;
2. remix gently when appropriate;
3. allow bubbles to disappear;
4. reposition or replace the tube; and
5. repeat the measurement.

Do not delete an individual reading without recording a technical reason.

9.6.2 Independently prepared assay replicates

Independent replicates require separate pipetting of sample and reagent.

When replicate concentrations disagree:

1. check sample identity, calculations, pipetting, mixing, incubation, and tube placement;
2. remeasure the existing tubes;
3. prepare fresh independent replicates if disagreement remains; and
4. reject the result when no reproducible value can be obtained.

Near zero concentration, relative standard deviation may be misleading. Evaluate the absolute spread together with LOD and LOQ.

9.7 Evaluate blank and QC samples

9.7.1 Reagent blank

The blank must:

- use the same reagent and vessel type as the samples;
- contain no intentionally added dsDNA;
- be prepared and incubated with the batch; and
- remain within the laboratory-defined blank acceptance criterion.

An elevated or unstable blank may indicate contamination, carryover, bubbles, degraded reagent, an unsuitable calibration, or an optical problem.

When the blank fails:

1. prepare and measure a fresh blank;
2. inspect reagent, tubes, pipettes, and the optical chamber;
3. replace contaminated or degraded materials; and
4. repeat the calibration or affected batch when necessary.

Do not use a clearly invalid blank to correct unknown samples.

9.7.2 Quality-control samples

A QC sample passes only when its result lies within a predefined acceptance interval.

The interval must be established before reviewing the batch and may be based on:

- an independently assigned target value;
- historical control data;
- an approved recovery range; or
- another documented laboratory criterion.

Do not create or widen QC limits after seeing the result.

When QC fails:

1. stop accepting unknown-sample results;
2. check the calibration, blank, dilution, pipetting, mixing, reagent, and tube placement;
3. repeat the QC measurement after correcting an identified technical problem;
4. prepare a fresh QC assay when required; and
5. repeat affected unknown samples if QC performance cannot be restored.

Results measured since the last passing QC sample are potentially invalid.

9.8 Interpret concentration, prediction error, LOD, and LOQ

9.8.1 Concentration basis

The current firmware reports the principal concentration on the original-sample basis defined by the calibration. The corresponding in-assay concentration may also be retained in the CSV output.

Do not mix original-sample and in-assay values.

The normal assay dilution caused by adding 1 µL sample to 200 µL reagent is already represented by the calibration metadata. Do not multiply the displayed original-sample result by 201.

Apply an additional factor only when the original unknown sample was deliberately diluted before assay preparation:

[c_original=c_reported×F_pre-dilution]

Record the pre-dilution factor.

9.8.2 Prediction error

For an imported multipoint calibration, the displayed `95% PI +/-` value is the half-width of the two-sided prediction interval for one analytical determination.

It is not the same as:

- the standard deviation of the three sensor readings;
- variation between independently prepared replicates; or
- uncertainty introduced by a manual pre-dilution.

Report concentration with its prediction error only when:

- a valid multipoint calibration is active;
- the prediction interval is available;
- the result lies within the calibration range;
- the result is at or above the LOQ; and
- blank, QC, and replicate criteria have passed.

9.8.3 LOD and LOQ

Use the LOD and LOQ belonging to the active calibration and confirm their concentration basis.

Result	Interpretation	Reporting action
Below LOD	The signal cannot be distinguished reliably from the blank	Report `< LOD` or `not detected`, according to the laboratory procedure
At or above LOD but below LOQ	dsDNA may be detected, but the concentration is not reliably quantifiable	Report `< LOQ` or `detected, not quantified`
At or above LOQ and within range	The result may be quantified if all other criteria pass	Report concentration and prediction error where available
Above the upper calibration limit	The result is outside the validated range	Dilute the original sample and repeat the complete assay
Below the lower validated range	The numerical estimate is outside the validated range	Do not report it as a normal quantitative result

Negative blank-corrected values must not be reported as negative dsDNA concentrations. Report them according to the laboratory procedure, normally as `< LOD`.

9.9 Accept, repeat, dilute, or reject

Observation	Decision	Required action
Calibration valid; blank and QC pass; replicates agree; result at or above LOQ and within range	**Accept**	Report the result and record any pre-dilution
One tube shows unstable readings, bubbles, contamination, or incorrect placement	**Repeat measurement**	Correct the technical problem and remeasure the same tube
Independent assay replicates remain inconsistent	**Repeat assay preparation**	Prepare fresh independent replicates
Calibration is incompatible, expired, or incomplete	**Reject**	Load or create an acceptable calibration
Result is above range or the detector response is saturated	**Dilute and repeat**	Dilute the original sample, repeat the complete assay, and apply the recorded pre-dilution factor
Result is below LOQ	**Do not accept as a normal quantitative result**	Report according to the LOD/LOQ category
Blank fails	**Hold the batch**	Investigate and obtain an acceptable blank
QC fails	**Hold or reject affected results**	Correct the cause and repeat QC and affected samples
Wrong sample, dilution, reagent, protocol, unit, or calibration was used	**Reject**	Repeat the assay correctly
No technically justified and reproducible result can be obtained	**Reject**	Record the result as invalid and document the reason

9.10 Minimum documentation

Record at least:

- date and operator;
- instrument and firmware version;
- Calibration Suite version;
- calibration ID, date, type, status, and range;
- calibration report and operational JSON filename;
- reagent and standard identification;
- blank result;
- QC target, acceptance interval, and result;
- sample identifier;
- individual replicate results;

- measurement-cycle mean and standard deviation;
- prediction error where available;
- LOD and LOQ with concentration basis;
- pre-dilution factor;
- final reported result;
- acceptance decision; and
- reason for every repetition, dilution, exclusion, or rejection.

Retain the original data. Repeated or corrected measurements must not overwrite earlier results without traceability.

9.11 Bench-side checklist

Before reporting a quantitative result, confirm:

- **Accepted calibration active?**
- **Calibration ID and range correct?**
- **Fresh blank passed?**
- **QC passed?**
- **Replicates consistent?**
- **Result at or above LOQ?**
- **Result within the calibration range?**
- **Prediction error available when required?**
- **Pre-dilution recorded?**
- **No unresolved technical fault?**

Only when all applicable checks pass may the result be accepted as quantitative.

10 DIY-QuantiFluorONE-dsDNA-Fluorometer – Validation and Analytical Performance

Operational calibration status: ACTIVE
Method performance status: VERIFIED
Operational range: 0–400 ng/µL dsDNA
Intended use: Research, teaching, and workshop measurements

10.1 Scope

The DIY fluorometer was evaluated under the final optical configuration and the standard QuantiFluor® ONE workflow of 1 µL sample plus 200 µL reagent.

Routine operation uses a two-point calibration consisting of a reagent blank and a nominal 400 ng/µL lambda-DNA standard. Analytical performance was assessed by comparison with a Promega Quantus Fluorometer, blank measurements, and the three consecutive TSL2591 readings recorded during each measurement cycle.

The Quantus was treated as a commercial comparison instrument rather than as a traceable reference method. It was used as an **independent external comparator**, not as a higher-ranked analytical truth.

10.2 Calibration and comparison design

Two nominal 400 ng/µL lambda-DNA standards were prepared separately:

- one standard for the Quantus calibration;
- one standard for the DIY fluorometer calibration.

The DIY calibration standard returned 395 ng/µL when measured on the Quantus, corresponding to 98.75% recovery. The DIY calibration was not retrospectively changed from 400 to 395 ng/µL.

Fifteen paired values covered the blank-to-upper-standard interval. The 400 ng/µL DIY result is the upper two-point calibration anchor. It is shown in the figures but excluded from the independent linear regression.

10.3 Why the Quantus is not treated as analytically superior

The comparison is arranged with the Quantus concentration on the x-axis only because an external comparator is required to assess the output of the two-point-calibrated DIY instrument.

This does **not** imply that the Quantus is analytically privileged, traceable, or error-free. It simply provides an independent comparison scale. The resulting figures answer the practical question:

How closely do the concentrations reported by the two-point-calibrated DIY fluorometer follow an independently measured comparator across the tested range?

10.4 Linearity of the two-point-calibrated DIY results

The most direct practical linearity check is the comparison of the concentrations reported by the two-point-calibrated DIY fluorometer with the independently obtained Quantus concentrations.

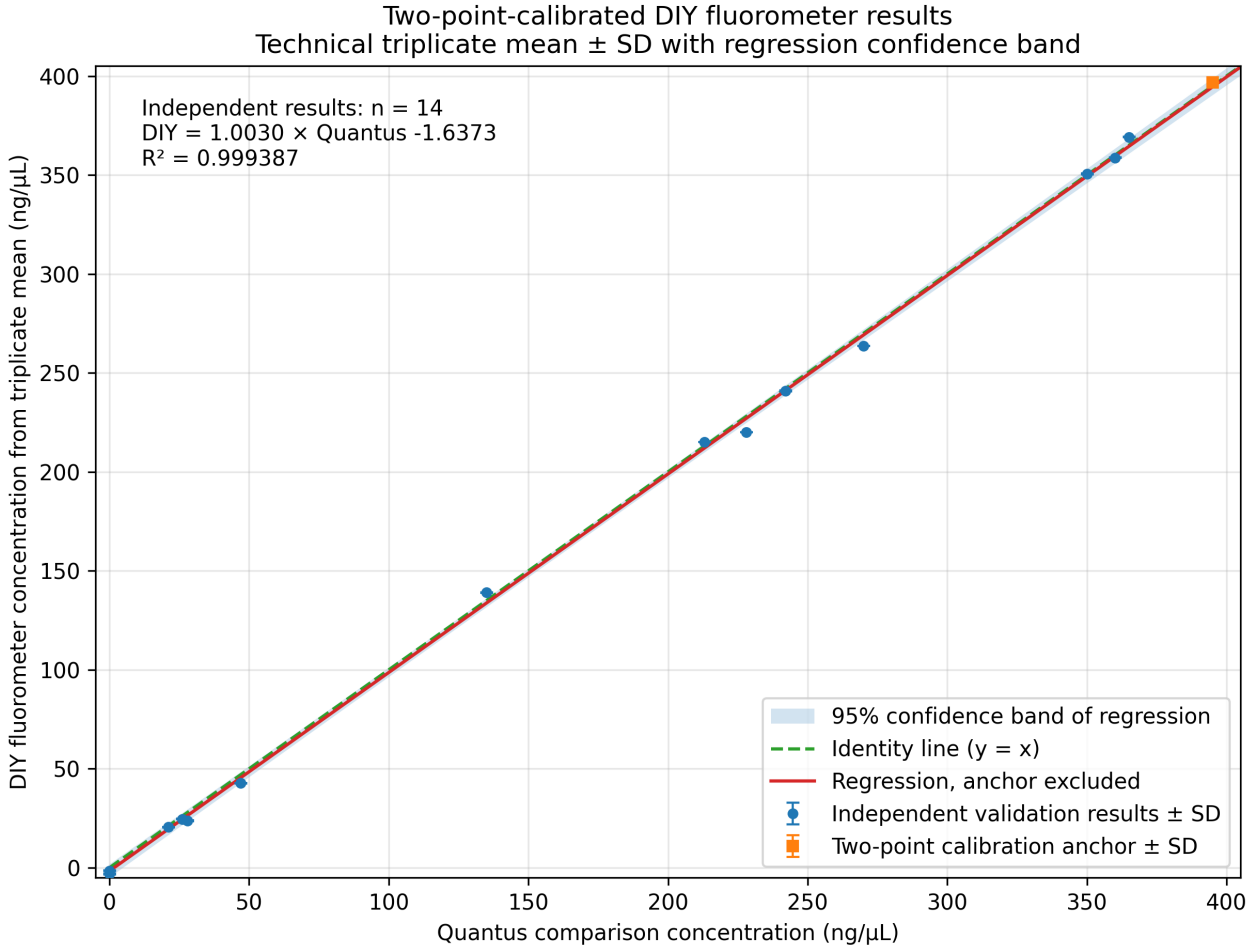

Figure 10.1: Two-point-calibrated DIY fluorometer results shown as technical triplicate mean ± SD. Marker size was reduced so that the error bars remain visible. The shaded band is the 95% confidence band of the independent linear regression. The square identifies the upper calibration anchor.

For the 14 independent results:

$$c_{\mathrm{DIY}} = 1.0030\, c_{\mathrm{Quantus}} - 1.6373$$

$$R^2 = 0.999387$$

The slope was close to 1, the intercept was close to 0, and no systematic curvature was evident over the tested interval.

The same technical-repeat data can also be visualized with 95% confidence intervals for the three technical readings:

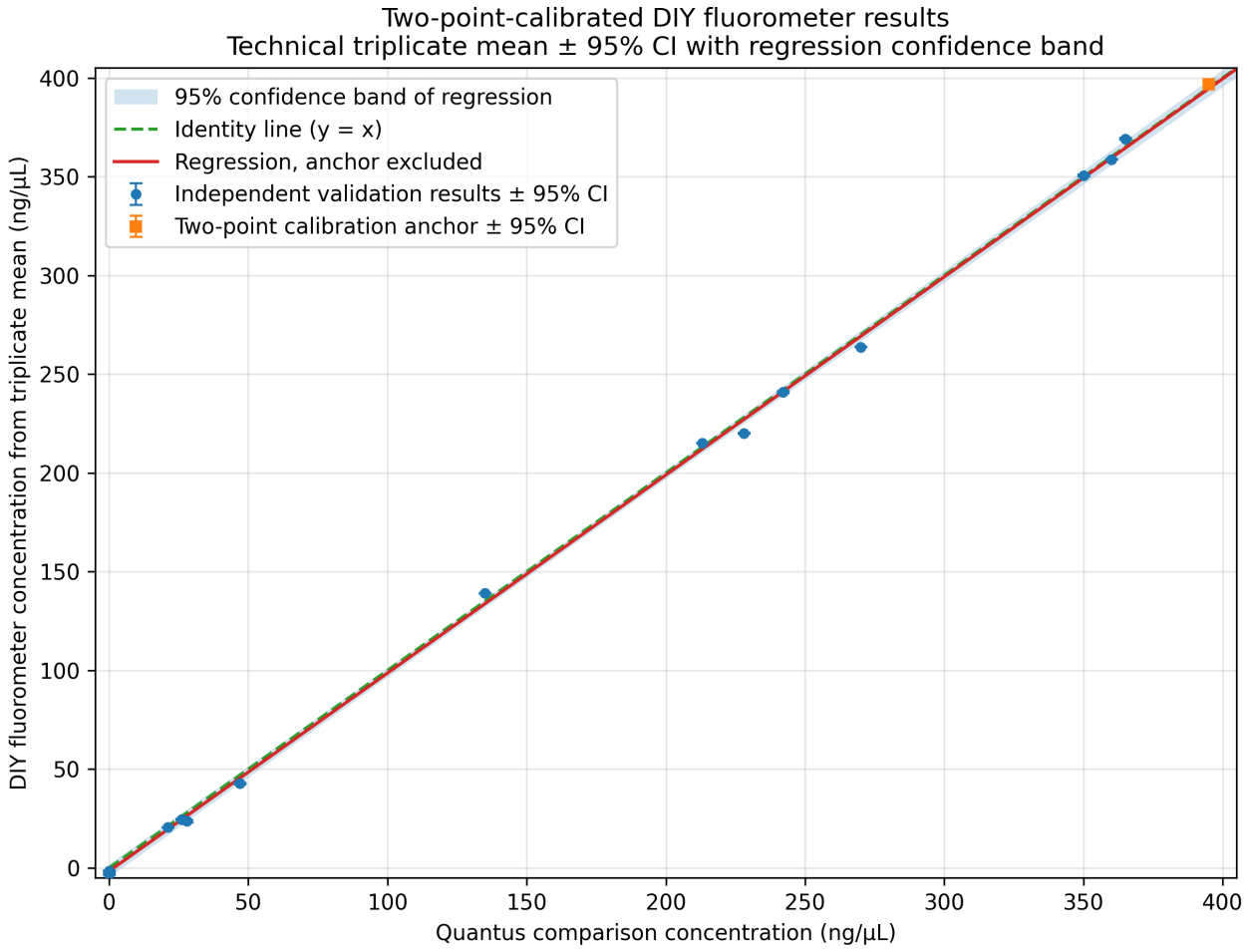

Figure 10.2: Two-point-calibrated DIY fluorometer results shown as technical triplicate mean ± 95% confidence interval. The shaded band is again the 95% confidence band of the independent regression. These intervals describe the uncertainty of the three consecutive sensor readings within one measurement cycle; they do not represent independently prepared assay replicates.

A formal curvature assessment such as a Mandel fitting test would be more defensible with independent replicate preparations per concentration level. In the present dataset, the triplicate readings are technical sensor repeats rather than independent assay replicates. They support the visual conclusion of linearity and short-term repeatability, but they should not be overinterpreted as full replicate-level evidence of non-linearity testing.

For transparency, including the upper calibration anchor gives:

$$c_{\mathrm{DIY}} = 1.0067\, c_{\mathrm{Quantus}} - 1.9467$$
$$R^2 = 0.999415$$

10.5 Agreement with the Quantus

A Bland–Altman plot is a useful complement to the linearity figure because it answers a different question: not *Is the relation linear?*, but *How large is the bias and how wide is the agreement interval between the two devices?*

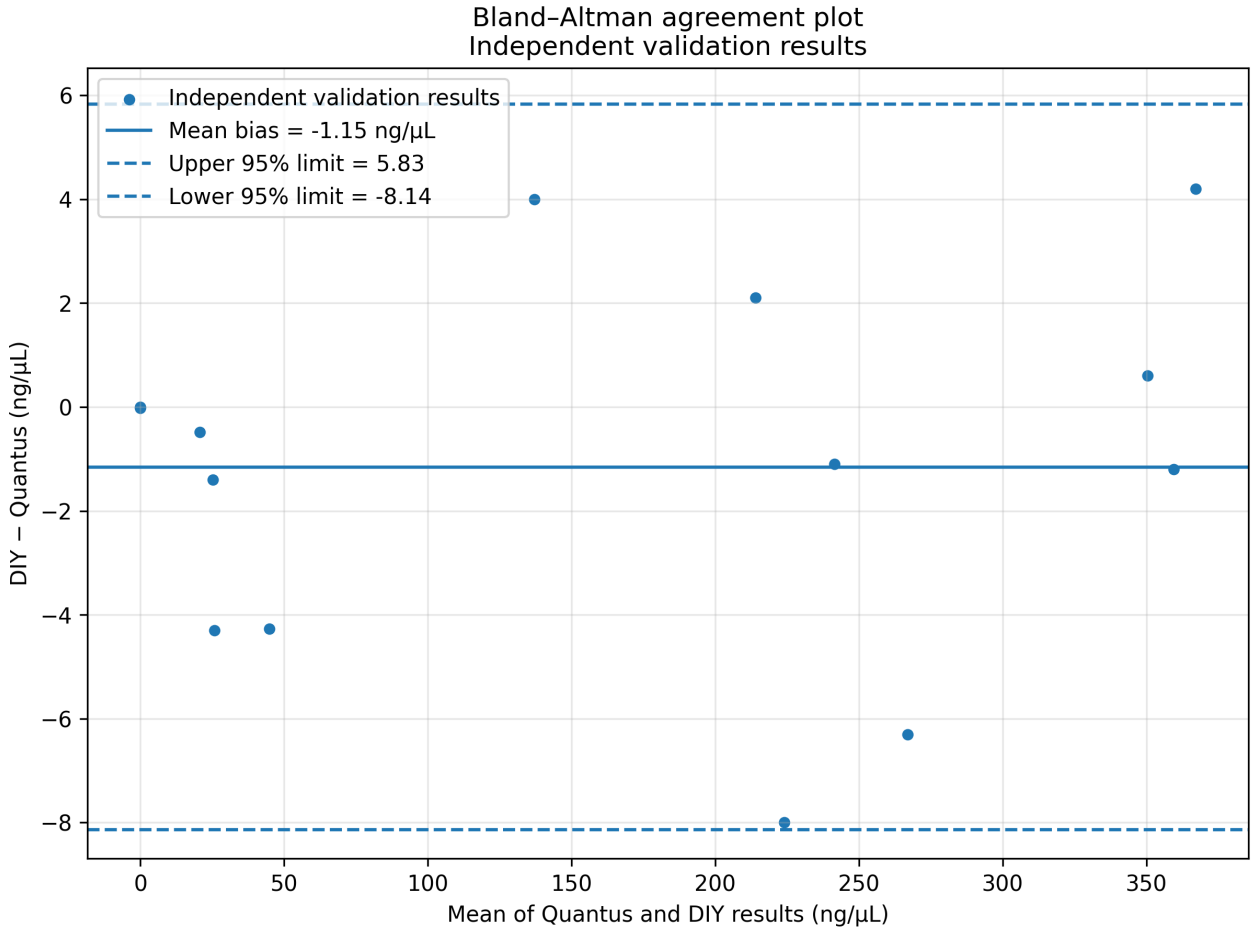

Figure 10.3: Bland–Altman agreement plot for the independent validation results. The solid line marks the mean bias, and the dashed lines mark the approximate 95% limits of agreement.

The independent comparison results gave:

Statistic	Result
Independent paired results	14
Mean bias, DIY − Quantus	-1.15 ng/µL
Median bias, DIY − Quantus	-0.79 ng/µL
Mean absolute error	2.71 ng/µL
Root mean squared error	3.62 ng/µL
Mean absolute relative difference above the primary LOQ	3.79%
Approximate 95% limits of agreement	-8.14 to 5.83 ng/µL

The Bland–Altman plot is therefore useful and appropriate here, but as a complement to the linearity assessment rather than a replacement for it.

10.6 What an RFU-versus-calculated-concentration plot can and cannot show

A plot of RFU versus the concentration calculated from the same two-point equation is useful as an **internal arithmetic-consistency check**. In the present device log, the corresponding relationship for rows 11–24 is:

$$RFU = 102.672895\,c - 0.002265$$

with:

$$R^2 = 0.999999999995$$

This near-perfect linearity is expected because the concentration values are mathematically derived from the same RFU values and the same two-point calibration equation.

Therefore, an RFU-versus-calculated-concentration plot is valuable for checking that the firmware applied the calibration equation correctly, but it is **not independent evidence of analytical linearity**. Independent linearity must instead be judged against an external reference scale or assigned concentration levels.

10.7 Blank performance, LOD, and LOQ

The primary blank study used ten independently prepared reagent-blank tubes. Each tube was measured in one complete cycle of three consecutive sensor readings.

The blank-based evaluation follows the project framework already described in Chapter 3 and uses the practical decision workflow documented here. The reporting of LOD and LOQ is aligned with the analytical concepts underlying **DIN 32645** and the calibration/working-range framework described in **DIN 38402-A51**.

Primary blank result	Value
Independent blank tubes	10
Mean VIS signal	9,557.563 counts
SD of blank-cycle means	126.514 counts
LOD, original-sample basis	7.902 ng/µL
LOQ, original-sample basis	23.946 ng/µL

Primary blank result	Value
LOD, in-assay basis	0.03931 ng/µL
LOQ, in-assay basis	0.11913 ng/µL

Two additional independently prepared blanks were later measured in eleven repeated cycles each. They produced confirmatory estimates of 7.027 ng/µL for the LOD and 21.294 ng/µL for the LOQ. These 22 cycles represent repeated measurements of two blank preparations and are not counted as 22 independent blanks.

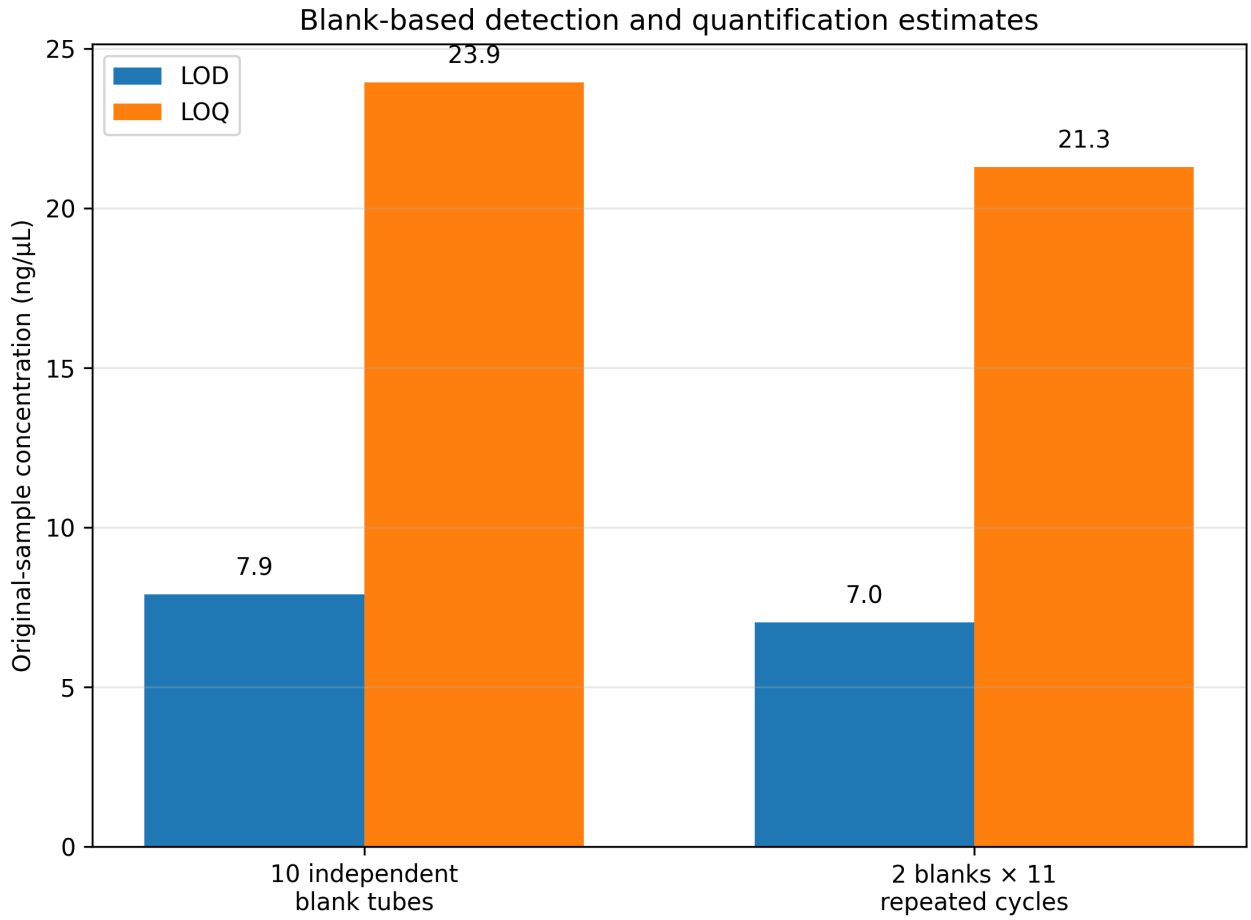

Figure 10.4: Primary and confirmatory blank-based LOD and LOQ estimates.

For concise operational use, the project uses approximately:

- **LOD: 8 ng/µL**
- **LOQ: 24 ng/µL**

10.8 Technical repeatability and sensor channels

Every measurement cycle contains three consecutive TSL2591 readings of the same assay tube.

In the final repeated-blank series:

- median within-cycle VIS CV: 0.033%;
- maximum within-cycle VIS CV: 0.348%.

These values demonstrate very good short-term sensor repeatability. They do not represent independent assay-preparation precision.

The analytical signal is calculated as:

$$VIS = FULL - IR$$

The IR channel is therefore included in every blank, RFU, and concentration calculation. In the evaluated blank data, IR subtraction slightly reduced rather than increased overall signal variability; the IR channel was not the dominant observed source of variation.

10.9 Operational range and reporting

The operational calibration range is **0–400 ng/µL**. LOD and LOQ define result categories within this range and do not remove the blank endpoint from the stated range.

Result	Reporting action
Blank or below approximately 8 ng/µL	Report as `< LOD` or not detected
Approximately 8 to below 24 ng/µL	Report as detected but below LOQ
Approximately 24 to 400 ng/µL	Report quantitatively when calibration, blank, and QC requirements pass
Above 400 ng/µL or sensor saturation	Dilute the original sample and repeat the assay

10.10 Routine quality control

Routine measurements continue to use an active two-point calibration. At least one independently prepared control standard should be measured before unknown samples are accepted.

A mid-range control is the most important routine check. Low- and high-level controls can be added where required.

The method-performance status and the current operational-calibration status must remain separate:

```
Operational calibration status: ACTIVE
Method performance status: VERIFIED
Operational range: 0-400 ng/µL
```

10.11 Conclusion

Under the tested configuration and assay conditions, the DIY fluorometer showed:

- a linear two-point-calibrated response over 0–400 ng/µL;
- close agreement with the Quantus comparison instrument;
- blank-based LOD and LOQ estimates of 7.9 and 23.9 ng/µL;
- very good short-term technical repeatability; and
- valid unsaturated measurements through the 400 ng/µL upper calibration endpoint.

The method is suitable for research, teaching, and workshop measurements. It is not intended for clinical diagnostic use.

10.12 Reproducibility files

The analysis source and data are stored under:

```
validation/
```

Additional processed data for the technical-triplicate concentration statistics are stored in:

```
validation/processed_data/diy_triplicate_concentration_summary.csv
```

11 Troubleshooting

This chapter summarizes the most common operational problems that may occur during assembly, firmware installation, calibration, and routine measurements. It is intended as a practical troubleshooting guide for workshop participants and laboratory users. More detailed background information is provided in the previous chapters.

11.1 Hardware

Symptom	Possible cause	Recommended action
Display remains dark	Battery discharged	Recharge the PyBadge or connect USB power.
Device does not start	CircuitPython not installed correctly	Repeat the firmware installation (Chapter 5).
LED does not illuminate	Loose cable or incorrect connector	Verify all wiring and connector orientation.
Sensor not detected	Qwiic/STEMMA cable disconnected or I²C scan unsuccessful	Check the cable, connectors, and I²C connections, then restart the instrument.
Very low fluorescence signal	Optical filters installed incorrectly	Verify the orientation and seating of both optical filters.
Signal permanently saturated	Light shield open, optical light leak, or excessive sample signal	Close the light shield and inspect the optical path. If the sample remains above range, dilute it and repeat the assay.

11.2 Firmware and Software

Symptom	Possible cause	Recommended action
Calibration cannot be loaded	Missing or incompatible JSON file	Export the calibration again using the Calibration Suite.
Calibration import fails	Invalid JSON format	Validate the calibration file before copying it to the device.
Device reports “uncalibrated”	No calibration selected	Load a valid calibration before measuring samples.
Device freezes during startup	Incomplete firmware installation	Reinstall the firmware according to Chapter 5.

11.3 Calibration

Symptom	Possible cause	Recommended action
Blank values unstable	Ambient light or contaminated PCR tube	Repeat the blank measurement using a clean tube under stable conditions.
Calibration rejected	Standards prepared incorrectly	Prepare fresh standards and repeat the calibration.
Unknown outside calibration range	Sample concentration exceeds the validated range	Dilute the sample and repeat the measurement.
High calibration uncertainty	Pipetting errors or degraded standards	Prepare fresh standards and recalibrate.

11.4 Measurement

Symptom	Possible cause	Recommended action
Large replicate variation	Air bubbles or insufficient mixing	Mix gently and remove visible bubbles before measurement.
Negative or below-zero calculated result	Sample signal below the stored blank or blank mismatch	Measure a fresh reagent blank and verify the calibration. Do not report a negative dsDNA concentration; apply the LOD/LOQ reporting rules from Chapter 9.
Unexpectedly low concentration	Pipetting error or degraded reagent	Verify pipetting accuracy and reagent quality.
Unexpectedly high concentration	Sample contamination	Prepare a fresh aliquot and repeat the measurement.
Measurement exceeds calibration range	Sample concentration above the validated range	Dilute the sample and repeat the analysis.

11.5 Calibration Suite

Symptom	Possible cause	Recommended action
CSV file cannot be imported	Incorrect or unsupported CSV structure	Check the delimiter, column headers, decimal format, and required calibration-data columns.
Calibration model unavailable	Insufficient calibration points	Acquire additional standards covering the required concentration range.
Exported JSON rejected	Calibration incomplete	Complete the calibration workflow before exporting.

11.6 Preventive Maintenance

To ensure reproducible analytical performance:

- keep the optical compartment clean and free of dust;
- inspect optical filters regularly for contamination or damage;
- use clean PCR tubes for every measurement;
- ensure that the selected power source is sufficiently charged before extended measurement sessions;
- periodically verify the instrument using a quality-control sample.

11.7 Quick Troubleshooting Guide

Problem	Section
Device does not start	Hardware
LED or sensor not working	Hardware
Calibration cannot be loaded	Firmware and Software
Calibration rejected	Calibration
Unexpected measurement results	Measurement
CSV or JSON problems	Calibration Suite

12 Workshop Review, Quality Check and Further Applications

This chapter concludes the workshop with a brief review of the practical exercises, a final functional quality check, and an outlook on possible future applications of the underlying modular fluorescence platform.

12.1 Functional Quality Check

Before completing the workshop, verify that the instrument performs the following tasks successfully.

Step	Expected result
Power on the instrument	Device starts normally
Perform a blank measurement	Stable blank signal
Measure a quality-control standard	Result within the expected acceptance range
Repeat the measurement	Comparable concentration and repeatability
Save the measurement	CSV file created successfully
Load a stored calibration	Calibration accepted without errors

Successful completion of these checks confirms correct basic instrument operation. Quantitative results should only be accepted when the calibration, blank, quality-control, range, and reporting criteria described in Chapter 9 are also satisfied.

12.2 Workshop Review

12.2.1 Instrument

- Why is fluorescence detected at 90° to the excitation beam?
- Why are excitation and emission filters required?
- Why is a blank measurement performed before sample analysis?

12.2.2 Calibration

- When should a sample be diluted before measurement?
- What indicates that a calibration should be repeated?
- Why should quality-control samples be measured regularly?

12.2.3 Practical Operation

- Which file stores the calibration?
- Which file stores the measurement results?
- Which file is transferred from the Calibration Suite to the fluorometer?
- Why should calibration and validation data always be archived?

12.3 Further Applications

The DIY-QuantiFluorONE-dsDNA-Fluorometer described in this booklet is configured and evaluated specifically for the Promega QuantiFluor® ONE dsDNA assay.

The underlying instrument concept is modular. By selecting an appropriate excitation source, optical filters, detector settings, sample geometry, and calibration procedure, the same general design approach can provide a basis for other fluorescence-based analytical applications.

The examples below summarize possible starting points for further instrument development. The excitation and emission wavelengths are indicative optical design parameters for the respective fluorescence methods; they do not imply that these applications have been validated with the present DIY-QuantiFluorONE configuration.

Example application	Excitation LED	Emission / detection	Typical application
dsDNA (QuantiFluor® ONE)	485 nm	530 nm	DNA quantification
Chlorophyll-a	450 nm	680 nm	Phytoplankton and algal biomass
CDOM	350 nm	425 nm	Colored dissolved organic matter
Ammonium (OPA derivatisation)	350 nm	425 nm	Nutrient analysis
PAHs (naphtalene methyl derivates)	275 nm	AS7331 UVA channel (315–410 nm)	SPE from seawater

These examples illustrate how the modular optical concept may be extended beyond dsDNA quantification. They do not constitute validated instrument configurations or complete analytical methods.

Future applications may require different LEDs, excitation and emission filters, detector settings, sample holders, firmware parameters, and calibration procedures. Each modified configuration should therefore be characterized and validated for its intended analytical application before quantitative results are reported. Further examples of community-developed applications using the ioRodeo Open Colorimeter platform are collected in the ioRodeo community applications database (ioRodeo n.d.).

12.4 Workshop Completed

Congratulations!

You have successfully completed the DIY-QuantiFluorONE-dsDNA-Fluorometer workshop.

Participants should now be able to:

- assemble the instrument;

- install the firmware and software;
- perform blank, calibration and routine measurements;
- evaluate measurement quality and analytical performance; and
- understand how the same modular platform can be adapted to many other fluorescence-based analytical methods.

We encourage participants to further develop the platform, evaluate additional fluorescence assays and contribute improvements to the open-source project.

13 References and Acknowledgements

This project combines open-source hardware, commercial laboratory reagents, scientific standards, manufacturer documentation and original engineering developments. This chapter records the principal sources and acknowledges the contributions on which the DIY-QuantiFluorONE-dsDNA-Fluorometer is based.

The complete machine-readable bibliography is maintained in `references/references.bib`. Additional provenance information is available in `hardware_sources.yml`, `source_register.csv` and `standards_register.md`.

13.1 Scientific and Analytical References

The theoretical background is based on standard works on fluorescence spectroscopy (Lakowicz 2006; Valeur and Berberan-Santos 2012) and on published fluorescence-based dsDNA quantification methods (Singer et al. 1997).

The assay procedure and commercial reference instrument are documented by Promega (Promega Corporation 2022, 2024).

The calibration and detection-limit framework refers to DIN 38402-51 and DIN 32645 (*DIN 38402-51:2017-05: German Standard Methods for the Examination of Water, Waste Water and Sludge–General Information–Part 51: Calibration of Analytical Methods–Linear Calibration (a 51)* 2017; *DIN 32645:2008-11: Chemical Analysis–Decision Limit, Detection Limit and Determination Limit Under Repeatability Conditions–Terms, Methods, Evaluation* 2008).

The standards are cited as methodological references. Formal conformity assessment requires access to licensed copies and a documented comparison between the implemented calculations and the normative text.

13.2 Open Hardware and Software Sources

13.2.1 ioRodeo

13.2.2 ioRodeo Open Colorimeter Plus and related instrumentation

The firmware and hardware concepts are derived in part from the ioRodeo Open Colorimeter Plus project (ioRodeo 2026b).

The installed LED driver is the ioRodeo fixed-current radial 16 mA LED board, revision `ver_0p1_rev_3` (ioRodeo 2026a). For integration into the optical module, the through-hole LED is mounted on the rear side of the board. The board design files, licence, and source record are retained in the repository for traceability.

ioRodeo has also documented open photometric and fluorometric approaches to DNA quantification. The UV Open Colorimeter was used for UV-photometric dsDNA determination (Long 2024), while the Open Colorimeter Plus was evaluated for fluorometric dsDNA quantification with the AccuGreen assay

(Long 2023). Subsequent Open Colorimeter Plus hardware and firmware developments, including the dual-sensor architecture, are documented separately (Long and Dickson 2025).

These sources provide useful technical context for the open measurement-chain concept used in this project without being reproduced in detail in the workshop chapters.

13.2.3 ioRodeo Fluorometer Tube Holder

The mechanical development of the optical sample holder also draws on the open ioRodeo fluorometer tube-holder project:

https://github.com/iorodeo/fluorometer_tube_holder

The repository provides FreeCAD design files for the fluorometer tube holder and is distributed under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence.

The DIY-QuantiFluorONE holder was adapted for the Promega E4941 thin-walled 0.5 mL PCR tube, the installed 8 × 8 × 1 mm optical filters, one TSL2591 sensor, and the project-specific 90° optical geometry. The resulting holder combines design concepts from DIYNAFLUOR and ioRodeo.

The ioRodeo tube-holder and Open Colorimeter platform also illustrate how related modular optical arrangements can be used for photometric, turbidimetric and fluorometric measurements.

A UV-photometric approach to dsDNA determination using the UV Open Colorimeter is described here:

https://blog.iorodeo.com/uv-dna-quantification/

A fluorometric dsDNA determination using AccuGreen and the Open Colorimeter Plus is described here:

https://blog.iorodeo.com/open-colorimeter-plus-dna-quantitation/

In this fluorometric configuration, two TSL2591 sensors are used. The later ioRodeo hardware description documents the detector arrangement as a 180° geometry. This differs from the DIY-QuantiFluorONE configuration, which uses one TSL2591 detector fixed at 90° to the excitation axis.

Later firmware and hardware developments of the Open Colorimeter Plus, including changes relevant to this optical configuration, are described here:

https://blog.iorodeo.com/open-colorimeter-plus-firmware-and-hardware-updates/

13.2.4 DIYNAFLUOR

The optical module is derived from the published DIYNAFLUOR fluorometer concept and its open repository (Anderson et al. 2024; Trau Lab 2026). It was modified for the installed LED board, filters, sensor geometry and PyBadge enclosure.

13.2.5 Adafruit and CircuitPython

The controller and detector architecture uses the Adafruit PyBadge, TSL2591 sensor and PCA9546 multiplexer (Adafruit Industries n.d.a, n.d.c, n.d.b). The documented firmware environment uses CircuitPython 9.1.1 (Adafruit Industries 2024).

13.2.6 Spectral Verification

The project-specific filter measurements were acquired using a DIY spectrometer based on PySpectrometer2 (Wright 2026). The recorded curves were normalized to their respective maxima and are used to verify spectral position and band shape rather than absolute transmittance.

13.3 Commercial Components and Manufacturer Information

The validated configuration uses the Promega QuantiFluor® ONE dsDNA System (Promega Corporation 2022), Neemoo Ex470BP-40 and Em532BP-40 band-pass filters (Neemoo n.d.), and a 485 nm high-brightness radial excitation LED.

Product names and supplier references are provided for reproducibility and do not imply endorsement.

13.4 Engineering Contributions

The optical module is derived from the open-source DIYNAFLUOR design and was extensively adapted for the DIY-QuantiFluorONE-dsDNA-Fluorometer.

The modified 3D-print parts and their mechanical integration were developed collaboratively by:

- Dipl.-Ing. Matthias Birkicht, project author; and
- Florian Bock, 3D-Haven, 27570 Bremerhaven, Germany.

This work included adaptation of the optical parts, integration of the installed filters and sensor, accommodation of the rear-mounted LED board, and mechanical integration with the PyBadge enclosure.

13.5 Repository Source Records

Detailed source and provenance records are maintained in:

- `references/references.bib`
- `references/hardware_sources.yml`
- `references/source_register.csv`
- `references/standards_register.md`

These records should be updated whenever a source, component revision, software version or standard reference changes. The complete project repository and current release files are available on GitHub (Birkicht 2026). Project repository: DIY-QuantiFluorONE-dsDNA-Fluorometer

13.6 Acknowledgements

The project gratefully acknowledges the developers and contributors of the ioRodeo Open Colorimeter Plus, DIYNAFLUOR, Adafruit CircuitPython, and PySpectrometer2 projects, as well as the manufacturers and standards organisations whose documentation supported the development and validation of the instrument.

Special thanks to Joanne Long of ioRodeo, who kindly donated a fluorometric cuvette holder for 0.2 mL tubes so that we could test the ioRodeo Colorimeter Plus for fluorometric measurements.

Special thanks are extended to Florian Bock of 3D-Haven for the collaborative development and adaptation of the mechanical parts.

The DIY-QuantiFluorONE-dsDNA-Fluorometer and this workshop documentation were prepared in the context of the 2026 eDNA workshop with Dr. Achim Meyer within the SHARK-Seq Project (2026–2027).

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14 Appendices

The appendices provide practical reference material supporting the construction, calibration and routine operation of the DIY-QuantiFluorONE-dsDNA-Fluorometer. They complement the main chapters and are intended as a quick reference rather than a tutorial.

14.1 Appendix A – File and Data Overview

The appendices provide practical reference material supporting the construction, calibration and routine operation of the DIY-QuantiFluorONE-dsDNA-Fluorometer. They complement the main chapters and are intended as a quick reference rather than a tutorial.

File	Purpose
`quantifluorone_calibration.json`	Stores the active saved calibration used by the firmware.
`quantifluorone_multipoint.json`	Contains the firmware-compatible multipoint calibration imported through `SELECT → LOAD MP CAL`.
`quantifluorone_log_v100rc2.csv`	Stores measurement data generated by the firmware.
`quantifluorone_config.json`	Stores hardware, measurement, sample, assay, and calibration settings.
`quantifluorone_state.json`	Stores blank and interface state used by the firmware.
Calibration input CSV	Contains calibration standards and measurement data evaluated in the Calibration Suite.
Validation data files	Store raw, processed, and statistical data used for analytical validation.

> **Note:** The file names listed here correspond to the documented QuantiFluorONE `QF1-1.0.0-rc2` firmware configuration.

14.2 Appendix B – Data Fields

14.2.1 Calibration Input CSV

A calibration dataset should contain at least:

- Sample or standard identifier
- Known concentration
- Replicate number
- Measured fluorescence signal
- Blank information (if applicable)

14.2.2 Measurement Log CSV

The QuantiFluorONE measurement log records the measurement and calibration information required for traceability. Relevant fields include:

- Date and time
- Sample identifier
- Raw sensor values
- Calculated concentration
- Replicate statistics
- Calibration identifier
- LOD and LOQ (where available)
- Error or warning status

14.2.3 Multipoint Calibration JSON

The operational `quantifluorone_multipoint.json` contains the calibration information required by the firmware, including the applicable model parameters, concentration range, LOD, LOQ, calibration identity, and associated statistical information supported by the RC2 import format.

The complete calibration record remains the Calibration Suite report together with the original calibration dataset. The operational JSON should therefore not be treated as the sole calibration record. The exact field names are defined by the documented RC2 firmware import format and the corresponding Calibration Suite export.

14.3 Appendix C – Workshop Checklists

14.3.1 Instrument Checklist

☐ PyBadge starts normally
☐ LED operates correctly
☐ TSL2591 detected
☐ Blank measurement successful
☐ Correct calibration loaded and status checked
☐ Quality-control standard accepted
☐ CSV logging operational

14.3.2 Measurement Checklist

☐ Correct reagent
☐ Correct PCR tube
☐ Blank measured
☐ Calibration valid
☐ Sample mixed
☐ No visible air bubbles
☐ Sample within calibration range
☐ Replicate agreement acceptable
☐ Measurement stored successfully
☐ Result at or above LOQ for quantitative reporting
☐ Pre-dilution factor recorded, if applicable

14.4 Appendix D – Abbreviations and Symbols

Abbreviation	Meaning
DIY	Do It Yourself
dsDNA	Double-stranded DNA
QC	Quality Control
LOD	Limit of Detection
LOQ	Limit of Quantification
SD	Standard Deviation
CI	Confidence Interval
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
I²C	Inter-Integrated Circuit
Ex	Excitation
Em	Emission
RFU	Relative Fluorescence Unit
FULL	TSL2591 full-spectrum channel
IR	TSL2591 infrared channel
VIS	Project-defined visible signal (`FULL - IR`)
PI	Prediction Interval
OLS	Ordinary Least Squares

14.4.1 Mathematical Symbols

Symbol	Meaning
a	Regression intercept
b	Regression slope
R	Correlation coefficient
R^2	Coefficient of determination
c	Concentration
x	Reference concentration or calibration variable
y	Analytical response (RFU)

14.5 Appendix E – Repository Structure

The repository separates hardware, software, analytical tools, validation data, references, and workshop documentation into dedicated top-level areas.

The principal project directories are:

The principal project directories are:

- `bom/` – machine-readable bill of materials and component information;
- `hardware/` – mechanical CAD models, PCB source and manufacturing files, wiring information, and hardware documentation;
- `software/` – PyBadge bootloader, CircuitPython, and the supported QuantiFluorONE firmware release;

- `tools/` – Calibration Suite and supporting analytical utilities;
- `validation/` – analytical validation datasets, processed data, and evaluation scripts;
- `docs/figures/` – figures used by the GitHub documentation and workshop booklet;
- `docs/markdown/` – GitHub-readable workshop course chapters;
- `docs/references/` – bibliography and project source/provenance records;
- `docs/source_files/` – supporting source material; and
- `docs/workshop_booklet/` – Quarto source files used to generate the printable workshop booklet.

The Markdown and Quarto documentation contain the same scientific and operational content but serve different presentation formats. The Markdown chapters provide a directly browsable GitHub course, while the Quarto sources generate the formatted workshop booklet and PDF.

This organisation keeps hardware, software, analytical tools, validation evidence, references, and documentation separate while maintaining traceability between the documented instrument and the files required to reproduce it.